Infinitely Many Primes $p = m^2 + n^2$ with $n < p^{0.1129}$

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Abstract

We prove that infinitely many primes $p = m^2 + n^2$ have $n < p^{0.1129}$, and that infinitely many primes satisfy $\{\sqrt{p}\} < p^{-0.2742}$. This sharpens the exponents 0.119 and -0.262 of Harman and Lewis (2001). Following their approach, we derive both statements from a lower bound of the correct order of magnitude for the number of Gaussian primes of norm between $X/2$ and X in every sector of angle $X^{-0.3871}$, and we prove this bound with Harman's sieve and their arithmetic information. The new ingredients are three groupings of terms that their sieve discards: a factorization of a two-dimensional sum for the linear estimate, and two applications of the bilinear estimate to four-dimensional sums, one of them after a further application of Buchstab's identity. These groupings give the grouped terms asymptotic formulas, so that fewer terms are discarded. In addition, we give every step of the sieve argument in full, filling in the details that Harman and Lewis present briefly and making several of their statements and discarded regions precise, and we evaluate the losses of the discarded sums by two numerical methods.

Keywords: Gaussian primes, narrow sectors, Harman's sieve, Hecke characters, fractional part of $\sqrt{p}$.
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1 Introduction

A prime $p \equiv 1 \pmod 4$ is the norm $p = m^2 + n^2$ of a Gaussian prime $m + in$. Landau asked whether $n = 1$ occurs for infinitely many primes, that is, whether infinitely many primes have the form $m^2 + 1$ [20]. The question is open. Iwaniec [13] showed that infinitely many integers $m^2 + 1$ have at most 2 prime factors. Primes $p = m^2 + n^2$ are known with n in sparse sets of arithmetic form: the squares [6], the squares of primes [11], and the integers $y^2 + 1$ [18]. These results restrict the form of n, not its size. A weaker form of Landau's question asks how small n can be taken for infinitely many primes $p = m^2 + n^2$. With $m > n > 0$, the condition $n < p^\theta$ places the Gaussian prime $m + in$ inside a sector of angle about $p^{\theta-1/2}$ next to the real axis. So the problem is one of Gaussian primes in narrow sectors.

Under the Riemann hypothesis for the Hecke L-functions of $\mathbb{Q}(i)$, Ankeny [1] and Kubilius [15] showed that infinitely many primes $p = m^2 + n^2$ have $n \ll \log p$. Without hypotheses, every sector of angle greater than $X^{-3/10+\varepsilon}$, for fixed $\varepsilon > 0$, contains the expected number of Gaussian primes of norm in $(X, 2X]$, by a result of Ricci [21] quoted in Huang et al. [12, §1]. This covers $n < p^\theta$ for every $\theta > 1/5$. For smaller θ, Coleman [5] obtained $n < p^{0.1631}$ with the Rosser–Iwaniec sieve in number fields. Harman and Lewis [10] used essentially the arithmetic information of Coleman with Harman's sieve [9], and obtained $n < p^{0.119}$. Among the results that hold for every sector, this bound has remained the best known [14, §1.2]. Harman et al. [8] apply the same sieve to sectors in which the norm also lies in a short interval. Results that hold for almost all sectors go much further. Huang et al. [12] prove an asymptotic formula for the number of Gaussian primes of norm in $(X, 2X]$ in almost all sectors of angle greater than $X^{-3/5+\varepsilon}$, for every fixed $\varepsilon > 0$. Järviniemi and Teräväinen [14] find products of exactly 2 Gaussian primes in almost all sectors of area $(\log X)^{15.1}$ in the disc of radius $X^{1/2}$.

If $p = m^2 + n^2$ with $n < p^\theta$, then

$$0 < \sqrt{p} - m = \frac{n^2}{\sqrt{p} + m} < p^{2\theta-1/2}. \tag{1}$$

Let $\{x\}$ denote the fractional part of x. Then $\{\sqrt{p}\} < p^{-(1/2-2\theta)}$ once $\theta < 1/4$. Harman and Lewis proved the bound $n < p^\theta$ with $\theta = 0.119$ [10, Theorem 1], and deduced $\{\sqrt{p}\} < p^{-0.262}$ for infinitely many primes as a corollary [10, Theorem 4]. Earlier, Balog [3] and Harman [7] had obtained the exponent $1/4 - \varepsilon$ for every fixed $\varepsilon > 0$. The exponent 0.262 has remained the best known [16]. No exponent above $1/2$ is possible in such a bound, since $|\sqrt{p} - k| \geqslant 1/(2\sqrt{p}+1)$ for the integer k nearest to $\sqrt{p}$.

Our main result is the following.

Theorem 1 *There are infinitely many primes $p = m^2 + n^2$ with m, n positive integers and $n < p^{0.1129}$. Consequently, there are infinitely many primes p with $\{\sqrt{p}\} < p^{-0.2742}$.*

Theorem 1 follows from a lower bound for the number of Gaussian primes in narrow sectors. Here and below, the Gaussian integers are the complex numbers $m + in$ with $m, n \in \mathbb{Z}$, which form the ring $\mathbb{Z}[i]$, and the norm of $m + in$ is $|m + in|^2 = m^2 + n^2$. The primes of $\mathbb{Z}[i]$ are the Gaussian primes. A non-zero Gaussian integer, and in particular a product of Gaussian integers, is identified with its associate of argument in $[0, \pi/2)$, and arguments are taken modulo $\pi/2$.

Theorem 2 *Let $\theta = 0.1129$. There is a constant $c > 0$ such that, for every sufficiently large X,*

$$\#\big\{\varpi \in \mathbb{Z}[i] : \varpi \text{ Gaussian prime},\ X/2 \leqslant |\varpi|^2 < X,\ \psi \leqslant \arg\varpi < \psi + X^{\theta-1/2}\big\} \geqslant \frac{cX^{1/2+\theta}}{\log X}, \tag{2}$$

which holds for every $\psi \in \mathbb{R}$.

The sector contains about $X^{1/2+\theta}/(\pi \log X)$ Gaussian primes of norm in $[X/2, X)$ on average over ψ, so the lower bound (2) has the correct order of magnitude.

Proof of Theorem 1 from Theorem 2. Let $\theta = 0.1129$, and apply (2) with $\psi = 0$ and $X = 2^k$ for large k. The Gaussian primes of argument 0 are the rational primes $\ell \equiv 3 \pmod 4$, and fewer than $X^{1/2}$ of them have norm $\ell^2 < X$. So the sector contains a Gaussian prime $\varpi = m + in$ with $m, n > 0$. Its norm $p = m^2 + n^2$ is a rational prime with $X/2 \leqslant p < X$. Since $\theta < 1/2$ and $p < X$,

$$n = \sqrt{p}\,\sin(\arg\varpi) < \sqrt{p}\,X^{\theta-1/2} < \sqrt{p}\,p^{\theta-1/2} = p^\theta.$$

The ranges $[X/2, X)$ for different k are disjoint. So the primes p are distinct for different k, and there are infinitely many of them. The bound $\{\sqrt{p}\} < p^{-0.2742}$ follows from (1), because $1/2 - 2\theta = 0.2742$. □

1.1 Outline of the method

The proof of Theorem 2 refines the argument of Harman and Lewis [10]. Let $\mathcal{A}$ be the set of Gaussian integers in the sector of (2) with norm in $[X/2, X)$, and let $\mathcal{B}$ be the set of all Gaussian integers of the quadrant with norm in $[X/2, X)$, so $\mathcal{A} \subset \mathcal{B}$. The number of Gaussian primes in $\mathcal{A}$ is the number of elements of $\mathcal{A}$ without prime factors of norm below $X^{1/2}$. Buchstab's identity (6) decomposes this count into sums of two kinds. Sums of the first kind have an asymptotic formula: they equal $2X^{\theta-1/2}/\pi$ times the corresponding sums for $\mathcal{B}$, up to a negligible error. Harman and Lewis derive these formulas from bounds for Hecke L-functions, through linear (Type I) and bilinear (Type II) estimates. A sum has such a formula when its primes can be combined into variables whose norms lie in the ranges that these estimates admit.

Sums of the second kind consist of non-negative terms that enter with the sign +, so a lower bound may omit them. They are discarded. Each discarded sum, taken over $\mathcal{B}$ and divided by the number of Gaussian primes in $\mathcal{B}$, tends to an integral (18) of Buchstab's function ω over a region of exponents. This integral is the loss of the sum, and the sum of the losses is the total loss $\Phi(\theta)$. The same decomposition holds for the number of Gaussian primes in $\mathcal{B}$. So, up to a negligible error, the number of Gaussian primes in $\mathcal{A}$ is at least $2X^{\theta-1/2}/\pi$ times the difference between the number of Gaussian primes in $\mathcal{B}$ and the discarded sums over $\mathcal{B}$. With the losses of these sums, this gives the lower bound

$$\#\{\varpi \in \mathcal{A} : \varpi \text{ Gaussian prime}\} \geqslant \frac{2X^{\theta-1/2}}{\pi}\,\#\{\varpi \in \mathcal{B} : \varpi \text{ Gaussian prime}\}\,\big(1 - \Phi(\theta) - o(1)\big). \tag{3}$$

So (2) holds, with θ in place of 0.1129, for every $\theta \in (1/9, 1/8)$ with $\Phi(\theta) < 1$. For their decomposition, Harman and Lewis state that the total loss at $\theta = 0.119$ is below 0.943 [10, p. 134].

Besides the prime number theorem in progressions modulo 4, only the arithmetic information is quoted without proof. It consists of the sector inequality, the mean-value estimate for Hecke characters, the prime number theorem with Hecke characters, and Lemmas 6 and 7 of Harman and Lewis. Lemmas 20 to 23 of Appendix A state them.

1.2 Contributions

We add three groupings of terms that Harman and Lewis discard. A grouping combines the primes of a sum into the variables of the linear or bilinear estimate, after a further application of Buchstab's identity (6) where needed, and so gives an asymptotic formula to a set of terms on which their decomposition has none.

- *A new factorization on a region* R. On a two-dimensional region R of terms pq, with p and q Gaussian primes, Harman and Lewis discard the whole sum. We apply Buchstab's identity, which introduces a third prime r, and then the linear estimate to the factorization $p \cdot (qr)$. Harman and Lewis apply the linear estimate after Buchstab's identity only to the factorization $(pq) \cdot r$. On R the product pq is too large for the linear estimate, while p and qr satisfy its conditions, because q is small.
- *The bilinear estimate with any set of primes on a region* G. The other two groupings act on terms $pqrst$ in which two applications of Buchstab's identity have introduced primes r and s, and t is the remaining factor, the cofactor. Harman and Lewis discard these terms over four-dimensional regions, except on a few subregions. The bilinear estimate needs a prime variable and further variables u and v of norm at least about $X^{1/2-\theta}$, and it applies at every level of sifting of t. On a region G, we apply it with s as the prime variable and with the product of a non-empty subset of $\{p, q, r\}$ as u. Harman and Lewis use the subset $\{p, r\}$, in one of their regions only.
- *A further Buchstab step on a region* L. On a region L, we apply Buchstab's identity once more, at a level below every fixed power of X, which introduces a prime s' with $|s'| \leqslant |s|$. The linear estimate for $pqrs$ then treats the sum sifted at this level, and the bilinear estimate with s' as its prime variable treats the sum over s'. Here u may contain s, which on G is the prime variable.

Section 3 defines R, G and L and places them in the decomposition.

Table 3 shows the effect of each grouping. With the regions derived from the lemmas of Section 6 (Remark 27), the decomposition of Harman and Lewis has total loss 1 at $\theta = 0.1167825$. This is just below the range from 0.117 to 0.118 in which, by Harman and Lewis [10, p. 134], the constant in their lower bound ceases to be positive. They add that they did not spend much time on the numerical calculations. Adding R moves the value 0.1167825 to 0.1141073, and adding G and L moves it to 0.1128554, so that the total loss is below 1 at $\theta = 0.1129$. The losses are computed in floating point by two methods (Section 10). At $\theta = 0.1129$ the total loss is 0.9990033. The gap $1 - \Phi(0.1129) = 9.97 \times 10^{-4}$ is more than 800 times each of the following: the difference between the totals of the methods, the error estimate of the second method, and the change of the total of the first method under refinement of its quadrature.

We also give in full several steps that Harman and Lewis [10] present briefly: the fundamental lemma invoked in their Lemmas 8, 9, 11 and 12, the proofs of their Lemmas 8 to 10, the removal of the conditions that couple the variables of their bilinear estimate, and the extension of their Lemma 14 that part 6 of their §5 uses. Lemma 3, Lemmas 4 and 5 with Corollary 6, Lemma 7 and Lemma 12 fill in these steps. Appendix E compares the two arguments item by item, including several statements and discarded regions that are made precise here.

1.3 Structure of the paper

Section 2 fixes the notation. Section 3 states the decomposition: identity (12), the ranges of Table 1, the groupings R, G and L, and the total loss (20). Section 4 states the linear and bilinear estimates, a fundamental lemma over $\mathbb{Z}[i]$, and asymptotic formulas for sifting functions with a variable level, and Section 5 removes the conditions that couple the variables of a sum. Section 6 treats the sums over two primes. Section 7 proves identity (12), checks the hypotheses of the lemmas on each range of Table 1, and defines the discarded

regions. Section 8 treats the four-dimensional sums on G and L. Section 9 turns the discarded sums into integrals and proves the lower bound, Lemma 16. Section 10 evaluates the integrals numerically and proves Theorem 2, and Section 11 concludes. Appendix A collects standard facts on Gaussian primes and derives the linear and bilinear estimates from the arithmetic information. Appendix B proves Lemmas 3, 4, 5 and 7, and Appendix C proves Lemma 15. Appendix D lists the inequalities between the parameters, and Appendix E compares the argument with that of Harman and Lewis.

Theorem 2 follows from Lemma 16 and Table 2. Lemma 16 rests on Lemmas 9 to 15, and Lemmas 9 to 14 rest on the results of Sections 4 and 5.

2 Notation

Throughout, $1/9 < \theta < 1/8$, the range in which the inequalities of Table 4 hold, and X is large in terms of θ and of the other fixed parameters. Following Harman and Lewis [10, p. 120], put

$$\sigma := \frac{1}{2} - \theta, \quad \tau := \frac{1}{2} + \theta, \quad \rho := \frac{1+22\theta}{12}, \quad \zeta := \frac{1+4\theta}{3}, \quad \nu := \frac{7+10\theta}{12}, \tag{4}$$

which satisfy $\nu - \sigma = \rho$ and $\nu - \sigma/2 = \zeta$. Put $W := X^{\sigma}$ and $\mu := \exp((\log X)^{4/5})$. The exponents τ, ζ, ρ and ν bound the variables of the linear estimate, Lemma 1, and W bounds from below the variables u and v of the bilinear estimate, Lemma 2. The number $W^{-1} = X^{\theta-1/2}$ is the angle of the sector in (2). The number μ exceeds every fixed power of $\log X$ and is smaller than every fixed positive power of X. The fundamental lemma, Lemma 3, sifts by the primes of norm below μ^4, and the prime variable of the bilinear estimate has norm at least $\mu^4/2$.

For a Gaussian integer n, $n \sim N$ means $N \leqslant |n|^2 < 2N$. From now on, p, q, r, s and ϖ denote Gaussian primes, and $d(n)$ is the number of divisors of n. A sequence a_n is *divisor-bounded* if $|a_n| \leqslant d(n)^{C_0}$ for a fixed C_0. Fix $\psi \in \mathbb{R}$. The sets $\mathcal{A}$ and $\mathcal{B}$ of Section 1.1 are

$$\mathcal{A} := \{n \in \mathbb{Z}[i] : X/2 \leqslant |n|^2 < X,\ \psi \leqslant \arg n < \psi + W^{-1}\}, \qquad \mathcal{B} := \{n \in \mathbb{Z}[i] : X/2 \leqslant |n|^2 < X\},$$

and write $\mathcal{E}_u := \{n \in \mathcal{E} : u \mid n\}$ for $\mathcal{E} \in \{\mathcal{A}, \mathcal{B}\}$.

Buchstab's identity classifies a Gaussian integer by its least prime factor, so it needs a total order on the Gaussian primes. The norm does not give one, since a rational prime $\ell \equiv 1 \pmod 4$ is the norm of 2 Gaussian primes. As in Harman and Lewis [10, p. 122], the Gaussian primes are ordered by norm, and primes of equal norm by argument: $\varpi \prec \varpi'$ means $|\varpi| < |\varpi'|$, or $|\varpi| = |\varpi'|$ and $\arg \varpi < \arg \varpi'$. For $z \geqslant 1$ and a prime q, the sifting functions of Harman and Lewis [10, p. 121] are

$$S(\mathcal{E}, z) := \#\{n \in \mathcal{E} : |\varpi|^2 \geqslant z \text{ for all } \varpi \mid n\}, \qquad S(\mathcal{E}, q) := \#\{n \in \mathcal{E} : \varpi \succcurlyeq q \text{ for all } \varpi \mid n\},$$

and z and q are their *levels*. Using the letter S for both functions is a slight abuse of notation, but it causes no ambiguity: the level is a real number in the first and a prime in the second, and in both it excludes the prime factors below it, in norm or for $\prec$. The function $S(\mathcal{E}_u, z)$ also sifts the prime factors of u. For $n \in \mathcal{E}_u$, the quotient n/u is the *cofactor* of n. We also sift the cofactor alone:

$$S'(\mathcal{E}_u, z) := \#\{n \in \mathcal{E}_u : |\varpi|^2 \geqslant z \text{ for all } \varpi \mid n/u\}, \tag{5}$$

and $S'(\mathcal{E}_u, q)$ is defined in the same way with the condition $\varpi \succcurlyeq q$. Thus $S(\mathcal{E}_u, z) = S'(\mathcal{E}_u, z)$ if every prime factor of u has norm at least z, and $S(\mathcal{E}_u, z) = 0$ otherwise. In the same way, $S(\mathcal{E}_u, q) = S'(\mathcal{E}_u, q)$ if every prime factor ϖ of u satisfies $\varpi \succcurlyeq q$, and $S(\mathcal{E}_u, q) = 0$ otherwise. Classifying n/u by its least prime factor for $\prec$ gives Buchstab's identities

$$S'(\mathcal{E}_u, z) = S'(\mathcal{E}_u, w) - \sum_{w \leqslant |\varpi|^2 < z} S'(\mathcal{E}_{u\varpi}, \varpi), \qquad S'(\mathcal{E}_u, q) = S'(\mathcal{E}_u, w) - \sum_{\varpi \prec q,\ |\varpi|^2 \geqslant w} S'(\mathcal{E}_{u\varpi}, \varpi), \tag{6}$$

for $1 \leqslant w \leqslant z$ and for primes q with $|q|^2 \geqslant w$. These are Harman and Lewis [10, (13)], and the same identities hold for S in place of S' when $u = 1$.

The density of $\mathcal{A}$ in $\mathcal{B}$ is $2/(\pi W)$. A *sum* is a weighted count $\Sigma(\mathcal{E})$ of the elements of $\mathcal{E} \in \{\mathcal{A}, \mathcal{B}\}$, in which each element is counted with a real weight that does not depend on $\mathcal{E}$. For example, $|\mathcal{E}_u|$ gives the weight 1 to the multiples of u and 0 to the other elements, and $S'(\mathcal{E}_u, z)$ gives the weight 1 to the multiples of u whose cofactor has no prime factor of norm below z. A linear combination of sums is again a sum. A sum *has an asymptotic formula* if, for every $A > 0$,

$$\Sigma(\mathcal{A}) = \frac{2}{\pi W}\,\Sigma(\mathcal{B}) + \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big), \tag{7}$$

where the implied constant may depend on A and on the sum. A sum has an asymptotic formula *uniformly* in a family when (7) holds with the same implied constant for every member. Then, for a fixed C, a sum of at most $(\log X)^C$ members, each with the sign + or −, has an asymptotic formula, since A is arbitrary.

Throughout, $\varepsilon > 0$ denotes a fixed small margin. An inequality between exponents holds *with margin* ε if it still holds after ε is added to its smaller side. Implied constants depend only on A, on θ, on ε, on the exponent C_0 of the divisor bounds and on the number of variables. They never depend on dyadic boxes, intervals, levels or coefficients, which is the convention of Harman and Lewis [10, p. 121]. The letter C denotes a positive constant with the same dependence, not necessarily the same at each occurrence.

The function z below bounds the levels that Lemmas 4 and 5 admit: a weighted sum of $S'(\mathcal{E}_n, l(n))$ over $|n|^2 \leqslant X^{\tau-\varepsilon}$ has an asymptotic formula when the level $l(n)$ is at most a constant times $z(|n|^2)$. For real $y \geqslant 1$, let $j(y) := -1$ if $y \geqslant W$, and otherwise let $j(y) \geqslant 0$ be the integer with $WX^{-(j(y)+1)\theta} \leqslant y < WX^{-j(y)\theta}$. Following Harman and Lewis [10, p. 128], put

$$z(y) := \Big(\frac{X^\tau}{y}\Big)^{1/(j(y)+2)}. \tag{8}$$

Thus $z(y) = X^\tau/y$ for $y \geqslant W$. For $y < W$ and $j = j(y)$,

$$z(y) > \Big(\frac{X^\tau}{WX^{-j\theta}}\Big)^{1/(j+2)} = X^\theta. \tag{9}$$

3 The decomposition and the total loss

In the notation of Section 2, the number of Gaussian primes in $\mathcal{A}$ is the sifting function $S(\mathcal{A}, X^{1/2})$. This section states the decomposition of $S(\mathcal{A}, X^{1/2})$, the groupings R, G and L, and the total loss. Lemma 16 proves the lower bound (3) in the form (48), where $\Phi_\varepsilon(\theta)$ tends to the total loss $\Phi(\theta)$ of (20) as $\varepsilon \to 0$.

3.1 The linear and bilinear estimates in exponent form

The asymptotic formulas rest on the linear and bilinear estimates, Lemmas 1 and 2. The arithmetic information quoted from Harman and Lewis [10] enters the argument only through these two lemmas. Consider a sum over primes $\varpi_1, \dots, \varpi_K$ and a cofactor t with $\varpi_1 \cdots \varpi_K t \in \mathcal{E}$, where $|\varpi_i|^2$ is about X^{x_i}. For a set $\mathcal{S}$ of indices, put $x_\mathcal{S} := \sum_{i\in\mathcal{S}} x_i$. The linear estimate applies when t is free, that is, subject to no sieve condition. Through Lemma 3, it also applies when t has no prime factor of norm below μ^4. In exponent form, it needs $\sum_{i=1}^K x_i \leqslant \tau$, or, for a partition of the indices into sets $\mathcal{U}$ and $\mathcal{V}$,

$$x_\mathcal{U} \leqslant \zeta, \qquad x_\mathcal{V} \leqslant \rho, \qquad \sum_{i=1}^K x_i \leqslant \nu. \tag{10}$$

The bilinear estimate applies at every level of sifting of t, since t becomes part of one of its variables, whose coefficients may be any non-negative divisor-bounded sequence. It needs an index i_0 and a non-empty set $\mathcal{S}$ of other indices with

$$\sigma \leqslant x_\mathcal{S} \leqslant \tau - x_{i_0}. \tag{11}$$

Then the product of the primes of $\mathcal{S}$ has norm at least W, up to a constant factor. So does the quotient of $\varpi_1 \cdots \varpi_K t$ by this product and by ϖ_{i_0}. These numbers are the variables u and v of Lemma 2, and ϖ_{i_0} is its prime variable ϖ. Lemma 8 states the condition $\sum_{i=1}^K x_i \leqslant \tau$ and (10), with margin ε, as (P1) for a free

cofactor and as (P2) for a cofactor without prime factors of norm below μ^4. It states (11), up to constant factors in the norms, as (P3). Its condition (P4) covers a cofactor sifted at a level that depends on the product $\varpi_1 \cdots \varpi_K$, by Lemmas 4 and 5. Conditions such as an order constraint $q \prec p$ or a sieve condition that depends on a prime couple the variables of a sum. Lemma 7 reduces such a sum to sums over Cartesian products of short intervals, to which (P1) to (P4) apply.

3.2 Decomposition of the sifting function

The decomposition begins with two applications of Buchstab's identity (6). Let Π_1 be the set of primes p with $X^\theta \leqslant |p|^2 < X^{1/2}$ and $z(|p|^2) \leqslant |p|^2$, and let Π_2 be the set of the other primes with $X^\theta \leqslant |p|^2 < X^{1/2}$. For $\mathcal{E} \in \{\mathcal{A}, \mathcal{B}\}$,

$$S(\mathcal{E}, X^{1/2}) = S(\mathcal{E}, X^\theta) - \sum_{p \in \Pi_1} S\big(\mathcal{E}_p, z(|p|^2)\big) - \sum_{p \in \Pi_2} S(\mathcal{E}_p, p) + \sum_{p,q}{}^{*} S(\mathcal{E}_{pq}, q), \tag{12}$$

where $\sum^*$ runs over the pairs with $p \in \Pi_1$, $q \prec p$ and $|q|^2 \geqslant z(|p|^2)$. The first application of Buchstab's identity gives the terms $S(\mathcal{E}_p, p)$ with $X^\theta \leqslant |p|^2 < X^{1/2}$. Since $|p|^2 < z(|p|^2)$ for $p \in \Pi_2$, the terms with $p \in \Pi_2$ have asymptotic formulas. For $p \in \Pi_1$, the second application lowers the level of $S(\mathcal{E}_p, p)$ to $z(|p|^2)$ and subtracts the terms of $\sum^*$. Lemma 13 proves (12). It gives the terms other than $\sum^*$ asymptotic formulas by Corollary 6 and Lemmas 1, 4 and 5. The rest of the argument concerns $\sum^*$.

The lemmas of Section 6 treat a pair (p, q) of $\sum^*$ according to the sizes of $|p|^2$ and $|q|^2$ as powers of X. We write $|p|^2 = X^\alpha$, $|q|^2 = X^\beta$, $|r|^2 = X^\gamma$ and $|s|^2 = X^\delta$. For $0 < \alpha < 1/2$ let $\beta_0(\alpha) := \log z(X^\alpha)/\log X$, the lower limit of β in $\sum^*$, so that

$$\beta_0(\alpha) = \begin{cases} \tau - \alpha & \text{for } \alpha \geqslant \sigma, \\ (\tau - \alpha)/(j+2) & \text{for } \alpha < \sigma, \text{ where } j = j(X^\alpha). \end{cases}$$

Section 7 discards the pairs of $\sum^*$ with $|p|^2|q|^4 \geqslant X$, which have cofactor 1 and whose sum over $\mathcal{B}$ is $\mathcal{O}(X/(\log X)^2)$. The other pairs have exponents in

$$\mathsf{D} := \big\{(\alpha, \beta) : \theta \leqslant \alpha < 1/2,\ \beta_0(\alpha) \leqslant \alpha,\ \beta_0(\alpha) \leqslant \beta < \min(\alpha, (1-\alpha)/2)\big\}. \tag{13}$$

Following Harman and Lewis [10, §5], whose numbering we keep, we split D into Ranges 1 to 6, listed in Table 1, and each of them into parts, which are polytopes. Figure 1 shows the ranges at $\theta = 0.1129$, with the regions D_k where the whole sum is discarded and the region R of (15). On each part, $\sum^*$ receives one of three *treatments*: an asymptotic formula for the whole sum; identity (29) of Lemma 11 or identity (31) of Lemma 12; or a discard of the whole sum. Both identities apply Buchstab's identity (6) twice. For $|pq|^2 \geqslant W$, the level $X^\tau/|pq|^2$ is the largest, up to a constant factor, that Lemma 4 admits for a sum of $S(\mathcal{A}_{pq}, \cdot)$ over (p, q). For a single pair (p, q) with $|p|^2|q|^4 \geqslant X^\tau$, which makes this level at most $|q|^2$, both applications at the level $X^\tau/|pq|^2$ give

$$S(\mathcal{A}_{pq}, q) = S\Big(\mathcal{A}_{pq}, \frac{X^\tau}{|pq|^2}\Big) - \sum_{\substack{r \prec q \\ |pqr|^2 \geqslant X^\tau}} S\Big(\mathcal{A}_{pqr}, \frac{X^\tau}{|pq|^2}\Big) + \sum_{\substack{s \prec r \prec q \\ |pqs|^2 \geqslant X^\tau}} S(\mathcal{A}_{pqrs}, s). \tag{14}$$

Summing (14) over a box $p \sim P$, $q \sim Q$ with $PQ \geqslant W$, with PQ in place of $|pq|^2$, gives identity (29). For the boxes with $PQ < W$, the level $X^\tau/(PQ)$ exceeds $z(PQ)$, and identity (31) is built in the same way at levels given by z. Where an identity applies, its sums over (p, q) and over (p, q, r) have asymptotic formulas. Its *third sum*, of $S(\mathcal{A}_{pqrs}, s)$ over primes $s \prec r \prec q \prec p$, has non-negative terms and the sign $+$, and it is discarded over a four-dimensional region of exponents $(\alpha, \beta, \gamma, \delta)$. Section 7 defines the discarded regions D_k and E_k of Table 1 and checks the hypotheses of the lemmas on each part, with the inequalities of Appendix D.

Range	Exponents	Treatment of $\sum^*$	Discarded	Third sums
1	$\sigma \leqslant \alpha < 1/2$	asymptotic formula (Lemma 9)	D_1	—
2	$1/3 \leqslant \alpha < \sigma$	identity (29), case (a), and case (b) on R (Lemma 11)	D_2	E_3, E_R
3	$\rho \leqslant \alpha < 1/3$	identity (29), case (a) (Lemma 11)	D_4	E_5
4	$\sigma - \theta \leqslant \alpha < \rho$	identity (29), case (a) (Lemma 11)	D_6	E_7
5	$\alpha < \sigma - \theta$, $\sigma \leqslant \alpha + \beta$	$S_<$, and identity (29), case (a) (Lemma 11)	D_8	E_9
5	$\alpha < \sigma - \theta \leqslant \alpha + \beta < \sigma$	$S_<$, and identity (31) with $j = 0$ (Lemma 12)	—	E_{10}
6	$\alpha + \beta < \sigma - \theta$	identity (31) with $j = 1$ (Lemma 12)	—	E_{11}

Table 1: The ranges of the domain D of (13), with Range 5 in 2 rows. For each range: the exponents of the pairs (p, q) of $\sum^*$; the treatment, where $S_<$ is the part of Lemma 11 or 12 with an asymptotic formula and the cases (a) and (b) are those of (28); the two-dimensional region where the whole sum is discarded; and the four-dimensional regions where the third sum of (29) or (31) is discarded. The regions are defined in Section 7, by (37)–(41) and (42). The index k of a region D_k or E_k is that of the loss φ_k of Harman and Lewis, and E_R has no counterpart there.

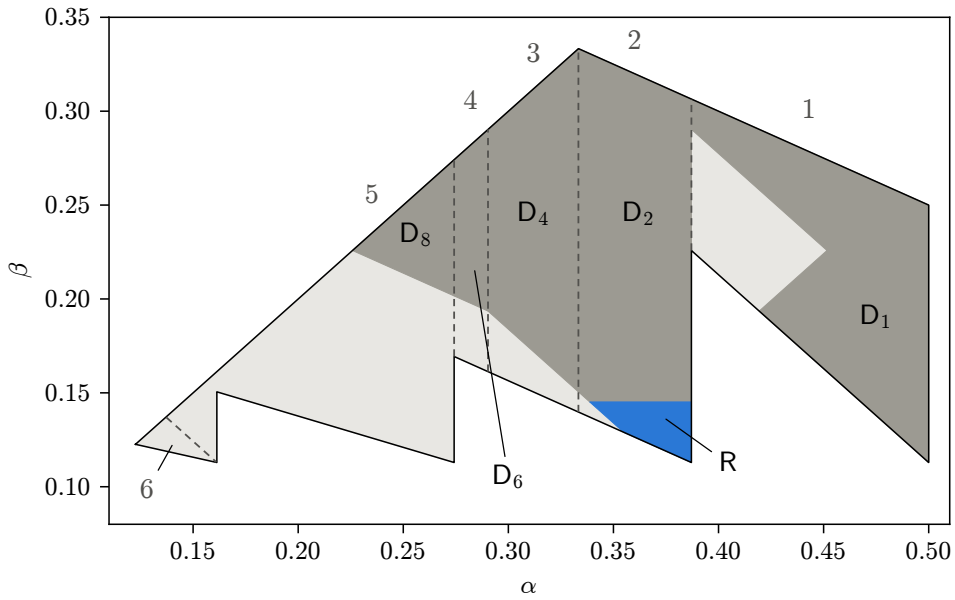


Figure 1: The domain D of (13) in the (α, β)-plane at $\theta = 0.1129$. Dashed lines separate the ranges of Table 1, numbered 1 to 6: the vertical ones are at $\alpha = \sigma = 0.3871$, $1/3$, $\rho = 0.29032$ and $\sigma - \theta = 0.2742$, and the oblique one is the line $\alpha + \beta = \sigma - \theta$ between Ranges 5 and 6. Dark gray: the regions D_1, D_2, D_4, D_6 and D_8 of Section 7. Blue: the region R of (15). Light gray: the rest of D.

3.3 Groupings on the regions R, G and L

Three groupings give asymptotic formulas to parts of the discarded sums. The grouping in Range 2 of the domain D of (13) (Table 1) acts on the region

$$\mathsf{R} := \big\{1/3 \leqslant \alpha < \sigma,\ \max(\zeta - \alpha, (\tau - \alpha)/2) \leqslant \beta \leqslant \rho/2\big\}. \tag{15}$$

Case (a) of (28) in Lemma 11 applies the linear estimate to the factorization $(pq) \cdot r$, which needs $\alpha + \beta \leqslant \zeta$. On R, where $\alpha + \beta \geqslant \zeta$, case (b) applies it to the factorization $p \cdot (qr)$, which needs $\alpha \leqslant \zeta$, $2\beta \leqslant \rho$ and $\alpha + 2\beta \leqslant \nu$. In Range 2, $\alpha < \sigma < \zeta$ and $\sigma + \rho = \nu$, so these conditions reduce to $2\beta \leqslant \rho$, the upper limit of β in (15). The lower limits of β there are $\zeta - \alpha$, below which case (a) applies, and $\beta_0(\alpha) = (\tau - \alpha)/2$. So on R, identity (29) replaces the discard of the whole sum, and only its third sum is discarded, over E_R.

The other two groupings give parts of the third sums of (29) and (31) asymptotic formulas. On a region G, the bilinear estimate applies to the products $pqrst$ with s as its prime variable: (11) holds with $x_{i_0} = \delta$ for a non-empty set $\mathcal{S} \subset \{p,q,r\}$. On a region L, Buchstab's identity (6) at level μ^4 writes $S(\mathcal{A}_{pqrs}, s)$ as the difference (45) of a sum sifted at level μ^4 and a sum over a further prime $s' \prec s$. The sum over s' enters with the sign $-$, so both sums need asymptotic formulas. The linear estimate treats the sum sifted at level μ^4 on the region I where the primes p,q,r,s satisfy $\alpha+\beta+\gamma+\delta \leqslant \tau$ or (10). Since $|s'|^2 \leqslant |s|^2$, the bilinear estimate with s' as its prime variable treats the sum over s' where (11) holds with $x_{i_0} = \delta$ for a non-empty set $\mathcal{S} \subset \{p,q,r,s\}$. With an index set written by its primes, as in $x_{\{p,q\}} = \alpha+\beta$, these regions are

$$\begin{aligned} \mathsf{G} &:= \bigcup_{\emptyset \neq \mathcal{S} \subset \{p,q,r\}} \big\{\sigma \leqslant x_{\mathcal{S}} \leqslant \tau - \delta\big\}, \\ \mathsf{I} &:= \big\{\alpha+\beta+\gamma+\delta \leqslant \tau\big\} \cup \bigcup_{\mathcal{U} \sqcup \mathcal{V} = \{p,q,r,s\}} \big\{x_{\mathcal{U}} \leqslant \zeta,\ x_{\mathcal{V}} \leqslant \rho,\ \alpha+\beta+\gamma+\delta \leqslant \nu\big\}, \\ \mathsf{L} &:= \mathsf{I} \cap \bigcup_{\emptyset \neq \mathcal{S} \subset \{p,q,r,s\}} \big\{\sigma \leqslant x_{\mathcal{S}} \leqslant \tau - \delta\big\}. \end{aligned} \tag{16}$$

Lemma 14 proves the formulas on G and L. These regions use the smallest prime of the product, s or s', as the prime variable of the bilinear estimate, and another prime in this role would not enlarge them. Indeed, let $x_0 \leqslant x_1$ be the exponents of the smallest prime of the product and of another prime ϖ of the product, and let (11) hold with $x_{i_0} = x_1$ for a set $\mathcal{S}$ of primes of the product with $\varpi \notin \mathcal{S}$. If $\mathcal{S}$ does not contain the smallest prime, then (11) holds with $x_{i_0} = x_0$ for the same $\mathcal{S}$. If it does, replacing the smallest prime in $\mathcal{S}$ by ϖ raises $x_{\mathcal{S}}$ by $x_1 - x_0$, to at least σ and at most $\tau - x_0$.

The region L contains the polytope

$$\mathsf{P} := \big\{\alpha+\beta \leqslant \sigma,\ \gamma+\delta \leqslant \rho,\ \alpha+\beta+\delta \geqslant \sigma,\ \alpha+\beta+2\delta \leqslant \tau\big\}. \tag{17}$$

On P, the partition $\mathcal{U} = \{p,q\}$, $\mathcal{V} = \{r,s\}$ satisfies (10), since $\sigma < \zeta$ and $\sigma + \rho = \nu$, and the set $\mathcal{S} = \{p,q,s\}$ satisfies (11) with $x_{i_0} = \delta$. Section 10 also evaluates the losses with P in place of L. Then L is replaced by a single polytope on which one partition and one set $\mathcal{S}$ serve every point, and the treatment of each four-dimensional box can be checked by hand.

3.4 The total loss

The lower bound of Lemma 16 drops the discarded sums over $\mathcal{A}$, and its main term is $2/(\pi W)$ times $S(\mathcal{B}, X^{1/2})$ minus the same sums over $\mathcal{B}$, as in (49). So each discarded sum is measured relative to $S(\mathcal{B}, X^{1/2})$. By Lemma 15, a discarded sum over $\mathcal{B}$ with exponents in a region, divided by $S(\mathcal{B}, X^{1/2})$, tends to the loss (18) of that region. For a two-dimensional region F and a four-dimensional region F', define the losses

$$\Phi(\mathsf{F}) := \iint_{\mathsf{F}} \omega\Big(\frac{1-\alpha-\beta}{\beta}\Big) \frac{d\beta\, d\alpha}{\alpha\beta^2}, \qquad \Phi(\mathsf{F}') := \iiiint_{\mathsf{F}'} \omega\Big(\frac{1-\alpha-\beta-\gamma-\delta}{\delta}\Big) \frac{d\delta\, d\gamma\, d\beta\, d\alpha}{\alpha\beta\gamma\delta^2}, \tag{18}$$

where ω is Buchstab's function, defined by $\omega(u) = 0$ for $u < 1$ and

$$u\omega(u) = 1 + \int_1^{u-1} \omega(t)\, dt \qquad (u \geqslant 1). \tag{19}$$

This equation defines ω implicitly, starting from $\omega(u) = 1/u$ for $1 \leqslant u \leqslant 2$, where the integral vanishes.

Let $\Phi(\theta)$ be the total loss

$$\Phi(\theta) := \sum_{k \in \{1,2,4,6,8\}} \Phi(\mathsf{D}_k) + \sum_{k \in \{3,\mathsf{R},5,7,9,10,11\}} \Phi\big(\mathsf{E}_k \setminus (\mathsf{G} \cup \mathsf{L})\big), \tag{20}$$

with D_k of (37), (38), (39), (40) and (41), E_k of (42), and G, L of (16). Let $\Phi_{\mathsf{P}}(\theta)$ be defined in the same way with P in place of L.

The total loss collects the losses of all sums that remain discarded: the sums over (p,q) on the regions D_k, and the third sums on the regions E_k outside $\mathsf{G} \cup \mathsf{L}$, where the groupings give asymptotic formulas. So $\Phi(\theta)$

measures the proportion of $S(\mathcal{B}, X^{1/2})$ that the discarded sums take up, and the lower bound of Lemma 16 is positive when $\Phi(\theta) < 1$.

Lemma 16 splits $\sum^*$ and the third sums into dyadic boxes. It applies Lemmas 9, 11, 12 and 14 to the boxes that lie inside a region where these lemmas apply, at distance at least ε from its boundary, and it bounds the other boxes by Lemma 15. Section 10 evaluates $\Phi(\theta)$ and shows $\Phi(0.1129) < 1$, which with Lemma 16 proves Theorem 2.

4 Asymptotic formulas from the linear and bilinear estimates

Each result of this section gives a sum an asymptotic formula in the sense of (7). Lemmas 1 and 2 below are the linear and bilinear estimates of Section 3, in the form used in this paper. Appendix A derives them from the arithmetic information of Harman and Lewis [10]. The other results of this section give asymptotic formulas for sums in which the cofactor is sifted. Lemma 3 sifts it at levels up to μ^4. Lemmas 4 and 5 sift the cofactors of the elements of $\mathcal{E}_n$ at a level $l(n)$ that depends on n. With Corollary 6, they extend Lemmas 8, 9 and 10 of Harman and Lewis [10] (Appendix E). Appendix B contains the proofs of Lemmas 3, 4 and 5.

Lemma 1 *Let $c_v \geqslant 0$ be divisor-bounded, and suppose that one of the following holds:*

(a) c_v is supported on $|v|^2 \leqslant X^\tau \mu^{-1}$;

(b) $c_v = \sum_{v=uu'} a_u b_{u'}$, where $a_u, b_{u'} \geqslant 0$ are divisor-bounded and supported on $u \sim U$, $u' \sim U'$, with $U \leqslant X^\zeta \mu^{-1}$, $U' \leqslant X^\rho \mu^{-1}$ and $UU' \leqslant X^\nu \mu^{-1}$.

Then $\sum_{v\neq 0} c_v |\mathcal{E}_v|$ has an asymptotic formula (7).

The bilinear estimate of Harman and Lewis [10, Lemma 5 and (30)] needs a prime variable. Its proof applies when the prime is restricted to any interval of norms above $\mu^4/2$ (Remark 25), and Section 5 restricts the prime to short intervals.

Lemma 2 *Let $C_1 \geqslant 1$, let $a_u, b_v \geqslant 0$ be divisor-bounded and supported on $u \sim U$ and $v \sim V$ with $U, V \geqslant W/C_1$, and let $\mu^4/2 \leqslant Z_1 < Z_2$. Then*

$$\sum_{u\sim U,\ v\sim V} a_u b_v \, \#\{\varpi : Z_1 \leqslant |\varpi|^2 < Z_2,\ uv\varpi \in \mathcal{E}\} \tag{21}$$

has an asymptotic formula, uniformly in U, V, Z_1 and Z_2.

Lemma 3 below is a fundamental lemma over $\mathbb{Z}[i]$. Every level at which it is applied is at most $\mu^4 = X^{o(1)}$, so Brun's pure sieve suffices (Appendix B).

Lemma 3 *Let $0 < \varepsilon_0 < 1$, let $c_u \geqslant 0$ be divisor-bounded, and let z_0 be a function on the support of c with $1 \leqslant z_0(u) \leqslant \mu^4$. Then $\sum_{u\neq 0} c_u S'(\mathcal{E}_u, z_0(u))$ has an asymptotic formula if one of the following holds:*

(a) c_u is supported on $|u|^2 \leqslant X^{\tau-\varepsilon_0}\mu^{-1}$;

(b) z_0 is constant, and $c_u = \sum_{u=u_1u_2} a_{u_1} b_{u_2}$, where $a_{u_1}, b_{u_2} \geqslant 0$ are divisor-bounded and supported on $u_1 \sim U$, $u_2 \sim U'$, with $U \leqslant X^\zeta \mu^{-1}$, $U' \leqslant X^{\rho-\varepsilon_0}\mu^{-1}$ and $UU' \leqslant X^{\nu-\varepsilon_0}\mu^{-1}$.

Lemma 3 sifts at levels up to μ^4, and the decomposition needs levels $l(n)$ up to $X^\tau/|n|^2$. Lemmas 4 and 5 reach them through the bilinear estimate. Buchstab's identity (6) lowers the level $l(n)$ to μ^4, where Lemma 3 applies, and subtracts a sum over a prime q with $\mu^4 \leqslant |q|^2 < l(n)$. Once q is confined to short intervals, n is the variable u of Lemma 2 in this sum, q is its prime variable, and the cofactor v of nq satisfies $|v|^2 > (X/2)/(|n|^2 l(n))$. So Lemma 2 applies when $|n|^2 \geqslant W/C_l$ and $l(n) \leqslant C_l X^\tau/|n|^2$, which are the hypotheses of Lemma 4.

Lemma 4 *Let $C_l \geqslant 1$. Let $a_n \geqslant 0$ be divisor-bounded and supported on $W/C_l \leqslant |n|^2 \leqslant X^{\tau-\varepsilon}$, and let l be any function on the support of a with $1 \leqslant l(n) \leqslant C_l X^\tau/|n|^2$. Then $\sum_{W/C_l\leqslant|n|^2\leqslant X^{\tau-\varepsilon}} a_n S'(\mathcal{E}_n, l(n))$ has an asymptotic formula, uniformly in a and l.*

For $|n|^2 < W$, the variable $u = n$ is too small for Lemma 2. The proof of Lemma 5 applies Buchstab's identity repeatedly, until the product of n and the primes it introduces has norm at least W, and then applies Lemma 4 to that product. The bound $l(n) \leqslant C_l\, z(|n|^2)$, with z of (8), ensures that each such product and its level satisfy the hypotheses of Lemma 4.

Lemma 5 *Let $C_l \geqslant 1$. Let $a_n \geqslant 0$ be divisor-bounded and supported on $1 \leqslant |n|^2 < W$, and let l be any function on the support of a with $1 \leqslant l(n) \leqslant C_l\, z(|n|^2)$, with z of (8). Then $\sum_{1\leqslant|n|^2<W} a_n S'(\mathcal{E}_n, l(n))$ has an asymptotic formula, uniformly in a and l.*

Corollary 6 *The sum $S(\mathcal{E}, X^\theta)$ of (12) has an asymptotic formula.*

Proof. The sum $S(\mathcal{E}, X^\theta) = S'(\mathcal{E}_1, X^\theta)$ is the case $a_n = \mathbf{1}[n = 1]$, $l \equiv X^\theta$ of Lemma 5. Its hypothesis holds, since $z(1) > X^\theta$ by (9). □

5 Removing the conditions that couple the variables

The sums of Harman's sieve run over products of primes with order constraints between them, conditions on products of their norms, and sieve conditions on the cofactor that depend on one of the primes. The bilinear estimate of Lemma 2 admits none of these conditions between its variables. This section removes them with short intervals. The order constraints and the conditions on products of norms are the cross-conditions of Merikoski [18, §7], and the short intervals give what he calls a finer-than-dyadic decomposition, which he builds from smooth weights [18, §3.1]. Heath-Brown and Li [11, §11.1] separate variables by intervals $(A, A + A/L]$, of the same shape as ours, and Harman and Lewis [10, p. 130] remove such conditions by Perron's formula (Appendix E). Every order constraint and every condition on a product of norms is constant on most Cartesian products of short intervals, and the sums over the other products are small. On a single Cartesian product, the sieve condition differs from sifting at a constant level only at the prime factors of the cofactor whose norms lie in a few short intervals. Lemma 7 reduces a sum with such conditions to sums over single Cartesian products of short intervals, and Lemma 8 gives these sums asymptotic formulas under the conditions of Section 3.1. The sieve conditions sift the cofactor as S' of (5) does, at a level that may depend on the primes. The other terms defined below are introduced here, for the bookkeeping of this reduction.

Let $H := (\log X)^B$ with B fixed, as in Lemma 17 of Appendix A, and put $\xi_\lambda := (1 + 1/H)^\lambda$ for integers $\lambda \geqslant 0$. The *short intervals* are $J_\lambda := [\xi_\lambda, \xi_{\lambda+1})$. For a prime q, let λ_q be the index with $|q|^2 \in J_{\lambda_q}$.

A *configuration* $\mathcal{C}$ consists of the following data:

(i) prime variables $\varpi_1, \dots, \varpi_k$, with k fixed and $|\varpi_i|^2 \in \mathcal{I}_i$ for intervals $\mathcal{I}_i \subset [\mu^4, X]$, the *ranges* of the variables. A *tuple* is a choice $\boldsymbol{\varpi} = (\varpi_1, \dots, \varpi_k)$ of these primes;

(ii) a cofactor t, which is a Gaussian integer;

(iii) finitely many *monomial constraints* c, each of the form $m_c(\boldsymbol{\varpi}) < Z_c$ or $m_c(\boldsymbol{\varpi}) \leqslant Z_c$, where $m_c(\boldsymbol{\varpi}) := \prod_{i=1}^k |\varpi_i|^{2e_{c,i}}$ with integers $|e_{c,i}| \leqslant 2$, not all 0, and $Z_c > 0$. We write $\mathbf{1}_c(\boldsymbol{\varpi})$ for the indicator that $\boldsymbol{\varpi}$ satisfies c. A lower bound $m_c(\boldsymbol{\varpi}) \geqslant Z_c$ is the constraint $m_c(\boldsymbol{\varpi})^{-1} \leqslant Z_c^{-1}$. An order constraint $\varpi_i \prec \varpi_{i'}$ is the monomial constraint with $m_c(\boldsymbol{\varpi}) = |\varpi_i|^2/|\varpi_{i'}|^2$ and $Z_c = 1$, in which the case $m_c(\boldsymbol{\varpi}) = 1$ is decided by $\prec$;

(iv) a *sieve condition* on t, with indicator $\Xi(\boldsymbol{\varpi}, t)$, of type (F), (C), (S) or (M): (F) none; (C) every prime factor of t has norm at least μ^4; (S) every prime factor ϖ' of t satisfies $\varpi' \succcurlyeq \varpi_{i_s}$, for a fixed index i_s; (M) every prime factor of t has norm at least $m_s(\boldsymbol{\varpi}) := Z_s \prod_{i=1}^k |\varpi_i|^{2f_i}$, with $Z_s > 0$ and reals $|f_i| \leqslant 2$ that do not depend on the tuple.

On every tuple that satisfies all the constraints, we require $\varpi_{i_s} \preccurlyeq \varpi_i$ for all i in type (S), and $\mu^4 \leqslant m_s(\boldsymbol{\varpi}) \leqslant \min_i |\varpi_i|^2$ in type (M). The subscript s of i_s, m_s and Z_s refers to the sieve condition, not to a prime s. The sum induced by $\mathcal{C}$ is

$$\Sigma_{\mathcal{C}}(\mathcal{E}) := \sum_{\varpi_1,\dots,\varpi_k} \sum_{t\in\mathbb{Z}[i]} \mathbf{1}[\varpi_1 \cdots \varpi_k t \in \mathcal{E}] \prod_{i=1}^k \mathbf{1}[|\varpi_i|^2 \in \mathcal{I}_i] \prod_c \mathbf{1}_c(\boldsymbol{\varpi})\, \Xi(\boldsymbol{\varpi}, t).$$

A configuration has the type of its sieve condition, and we identify it with the sum $\Sigma_{\mathcal{C}}$ that it induces. For example, for fixed primes p, q, r, s, the sum $S'(\mathcal{E}_{pqrs}, s)$ counts the cofactors t with $pqrst \in \mathcal{E}$ whose prime factors are all $\succcurlyeq s$. A sum of $S'(\mathcal{E}_{pqrs}, s)$ over primes in given ranges with $s \prec r \prec q \prec p$, such as a third sum of Section 3.2, is therefore a configuration with $k = 4$, the order constraints $s \prec r$, $r \prec q$ and $q \prec p$, and the sieve condition of type (S) with i_s the index of s.

For an index vector $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_k)$ of non-negative integers, let $\mathcal{T}(\boldsymbol{\lambda})$ be the set of tuples with $|\varpi_i|^2 \in \mathcal{I}_i \cap J_{\lambda_i}$ for all i. The *pieces* of $\mathcal{C}$ are the sums over a single set $\mathcal{T}(\boldsymbol{\lambda})$, without constraints, in which the cofactor is sifted at a constant level $y \geqslant 1$, or which have one further prime ϖ' with norm in a short interval $J_{\lambda'}$ and a free cofactor:

$$\mathcal{P}_{\boldsymbol{\lambda}}(\mathcal{E}; y) := \sum_{\boldsymbol{\varpi} \in \mathcal{T}(\boldsymbol{\lambda})} S'(\mathcal{E}_{\varpi_1 \cdots \varpi_k}, y), \qquad \mathcal{P}'_{\boldsymbol{\lambda},\lambda'}(\mathcal{E}) := \sum_{\boldsymbol{\varpi} \in \mathcal{T}(\boldsymbol{\lambda})} \sum_{|\varpi'|^2 \in J_{\lambda'}} |\mathcal{E}_{\varpi_1 \cdots \varpi_k \varpi'}|, \tag{22}$$

where ϖ' runs over primes and $\lambda' \geqslant 0$ is an integer, the index of a single short interval, not an index vector like $\boldsymbol{\lambda}$. For $y = 1$ the cofactor of $\mathcal{P}_{\boldsymbol{\lambda}}(\mathcal{E}; y)$ is free, since $S'(\mathcal{E}_u, 1) = |\mathcal{E}_u|$.

The index vector $\boldsymbol{\lambda}$ *satisfies* a constraint c if every tuple of $\mathcal{T}(\boldsymbol{\lambda})$ satisfies c, and *straddles* c if $\mathcal{T}(\boldsymbol{\lambda})$ contains tuples that satisfy c and tuples that do not. The *level* of $\boldsymbol{\lambda}$ is

$$y_{\boldsymbol{\lambda}} := \begin{cases} 1 & \text{in type (F)}, \\ \mu^4 & \text{in type (C)}, \\ \xi_{\lambda_{i_s}} & \text{in type (S)}, \\ \inf_{\mathcal{T}(\boldsymbol{\lambda})} m_s & \text{in type (M)}, \end{cases} \tag{23}$$

and its *window* is

$$\mathcal{W}(\boldsymbol{\lambda}) := \begin{cases} \emptyset & \text{in types (F) and (C)}, \\ \{\lambda_{i_s}\} & \text{in type (S)}, \\ \{\lambda' : J_{\lambda'} \cap [\inf_{\mathcal{T}(\boldsymbol{\lambda})} m_s, \sup_{\mathcal{T}(\boldsymbol{\lambda})} m_s) \neq \emptyset\} & \text{in type (M)}. \end{cases} \tag{24}$$

Since m_s varies by at most the factor $(1 + 1/H)^{2k}$ on $\mathcal{T}(\boldsymbol{\lambda})$, the window has at most $2k + 2$ elements. If $\boldsymbol{\lambda}$ satisfies all constraints, the requirements on types (S) and (M) give $y_{\boldsymbol{\lambda}} \leqslant \min_i |\varpi_i|^2$ on $\mathcal{T}(\boldsymbol{\lambda})$ and $\xi_{\lambda'} \geqslant \mu^4/2$ for $\lambda' \in \mathcal{W}(\boldsymbol{\lambda})$. On the tuples of such an $\boldsymbol{\lambda}$, the sieve condition $\Xi(\boldsymbol{\varpi}, t)$ and sifting at the constant level $y_{\boldsymbol{\lambda}}$ differ only at the prime factors of t with norm in $J_{\lambda'}$ for some $\lambda' \in \mathcal{W}(\boldsymbol{\lambda})$. So the part of $\Sigma_{\mathcal{C}}$ on the tuples of $\mathcal{T}(\boldsymbol{\lambda})$ differs from $\mathcal{P}_{\boldsymbol{\lambda}}(\mathcal{E}; y_{\boldsymbol{\lambda}})$ by at most the sum of $\mathcal{P}'_{\boldsymbol{\lambda},\lambda'}(\mathcal{E})$ over $\lambda' \in \mathcal{W}(\boldsymbol{\lambda})$.

Lemma 7 *Let $\mathcal{C}$ be a configuration, and suppose that the following pieces (22) have asymptotic formulas, uniformly in $\boldsymbol{\lambda}$ and λ':*

(a) $\mathcal{P}_{\boldsymbol{\lambda}}(\mathcal{E}; y_{\boldsymbol{\lambda}})$, with the level $y_{\boldsymbol{\lambda}}$ of (23), for every $\boldsymbol{\lambda}$ that satisfies all constraints;

(b) $\mathcal{P}_{\boldsymbol{\lambda}}(\mathcal{E}; 1)$ for every $\boldsymbol{\lambda}$ that straddles some constraint;

(c) $\mathcal{P}'_{\boldsymbol{\lambda},\lambda'}(\mathcal{E})$ for every $\boldsymbol{\lambda}$ that satisfies all constraints and every λ' in its window $\mathcal{W}(\boldsymbol{\lambda})$ of (24).

Then $\Sigma_{\mathcal{C}}$ has an asymptotic formula.

The pieces in (b) of Lemma 7 are the *straddling pieces* of $\mathcal{C}$, and the pieces in (c) are its *window pieces*. Summed over $\boldsymbol{\lambda}$ and λ', the straddling and window pieces are small over $\mathcal{B}$, and their asymptotic formulas carry this bound to $\mathcal{A}$. Appendix B proves Lemma 7.

The hypotheses of Lemma 7 are checked with Lemma 8 below. A piece has the primes $\varpi_1, \dots, \varpi_K$, with $K = k$ for $\mathcal{P}_{\boldsymbol{\lambda}}(\mathcal{E}; y)$, and with $K = k + 1$ and $\varpi_{k+1} = \varpi'$ for $\mathcal{P}'_{\boldsymbol{\lambda},\lambda'}(\mathcal{E})$. Its cofactor t is sifted at the constant level y, with $y = 1$ for $\mathcal{P}'_{\boldsymbol{\lambda},\lambda'}(\mathcal{E})$. The norm of ϖ_i lies in Λ_i, and the *exponents* of the piece are the x_i, where

$$\Lambda_i := \mathcal{I}_i \cap J_{\lambda_i} \quad (i \leqslant k), \qquad \Lambda_{k+1} := J_{\lambda'}, \qquad x_i := \frac{\log \inf \Lambda_i}{\log X}.$$

Each Λ_i lies in $[\mu^4/2, X]$ and has $\sup \Lambda_i \leqslant (1+1/H) \inf \Lambda_i$. So each $\log |\varpi_i|^2 / \log X$ lies within $\mathcal{O}(1/(H \log X))$ of x_i, and a condition that holds at (x_i) with margin ε holds with margin $\varepsilon/2$ for every tuple of the piece, once X is large. When the primes of a piece have names, an index set is written by its primes, as in $\mathcal{U} = \{p\}$, and i_0 by its prime, as in $\varpi_{i_0} = r$.

Lemma 8 *Let $C_1 \geqslant 1$ be fixed. A piece (22) has an asymptotic formula, uniformly in $\boldsymbol{\lambda}$ and λ', if one of the following holds.*

(P1) $y = 1$, *and either* $\sum_{i=1}^{K} x_i \leqslant \tau - \varepsilon$, *or the indices split as* $\mathcal{U} \sqcup \mathcal{V}$ *with*

$$x_{\mathcal{U}} \leqslant \zeta - \varepsilon, \qquad x_{\mathcal{V}} \leqslant \rho - \varepsilon, \qquad \sum_{i=1}^{K} x_i \leqslant \nu - \varepsilon.$$

(P2) $y = \mu^4$, *and the exponents satisfy one of the alternatives of (P1).*

(P3) There are an index i_0 and a non-empty set $\mathcal{S}$ of other indices such that

$$\inf \Big| \prod_{i \in \mathcal{S}} \varpi_i \Big|^2 \geqslant \frac{W}{C_1}, \qquad \frac{X/2}{\sup |\varpi_{i_0} \prod_{i \in \mathcal{S}} \varpi_i|^2} \geqslant \frac{W}{C_1},$$

with the infimum and the supremum over the tuples of the piece.

(P4) Every product $n = \varpi_1 \cdots \varpi_K$ of the piece satisfies

$$W/C_1 \leqslant |n|^2 \leqslant X^{\tau - \varepsilon}, \qquad y \leqslant C_1 X^{\tau}/|n|^2, \tag{25}$$

or

$$|n|^2 < W, \qquad y \leqslant C_1\, z(|n|^2). \tag{26}$$

Proof. In (P1), the piece is $\sum_{v \neq 0} a_v |\mathcal{E}_v|$, where a_v is the number of tuples of the piece with product v. This a_v is non-negative, at most $d(v)^K$, and supported on $|v|^2 \leqslant X^{\sum_{i=1}^{K} x_i + o(1)}$. When the indices split as $\mathcal{U} \sqcup \mathcal{V}$, $a_v = \sum_{v = uu'} a'_u b_{u'}$, where a'_u is the number of subtuples $(\varpi_i)_{i \in \mathcal{U}}$ with product u, and $b_{u'}$ is the number of subtuples $(\varpi_i)_{i \in \mathcal{V}}$ with product u'. Both a'_u and $b_{u'}$ are supported on at most 2 dyadic ranges. Lemma 1 applies, since the margin $\varepsilon/2$ absorbs the factor μ^{-1}.

In (P2), the piece is $\sum_{u \neq 0} a_u S'(\mathcal{E}_u, \mu^4)$, where a_u is the number of tuples of the piece with product u. Lemma 3 applies with $\varepsilon_0 = \varepsilon/4$: in case (a) when $\sum_{i=1}^{K} x_i \leqslant \tau - \varepsilon$, and in case (b) when the indices split as $\mathcal{U} \sqcup \mathcal{V}$, with u_1 and u_2 the products of the primes indexed by $\mathcal{U}$ and by $\mathcal{V}$, each in at most 2 dyadic ranges.

In (P3), the piece is

$$\sum_{u \neq 0,\ v \neq 0} a_u b_v \, \#\big\{ \varpi_{i_0} : |\varpi_{i_0}|^2 \in \Lambda_{i_0},\ uv\varpi_{i_0} \in \mathcal{E} \big\},$$

where a_u is the number of subtuples $(\varpi_i)_{i \in \mathcal{S}}$ with product u, and b_v is the number of pairs of a subtuple $(\varpi_i)_{i \notin \mathcal{S} \cup \{i_0\}}$ and a cofactor t sifted at level y with product v. Both are non-negative and divisor-bounded. The variable u lies in at most 2 dyadic ranges and v in at most 4. Both $|u|^2$ and $|v|^2$ are at least W/C_1. The norm of ϖ_{i_0} ranges over $\Lambda_{i_0} \subset [\mu^4/2, X]$, so Lemma 2 applies.

In (P4), the piece is $\sum_{n \neq 0} a_n S'(\mathcal{E}_n, y)$, where a_n counts the tuples with product n. Lemma 4 applies to the n that satisfy (25) and Lemma 5 to those that satisfy (26), both with the constant level y. □

6 Sums over two primes

The lemmas of this section treat the sum $\sum^*$ of (12) on dyadic boxes. They supply the treatments of Table 1, and Section 7 checks their hypotheses on each range.

Throughout, $p \sim P$ and $q \sim Q$, where P and Q are powers of 2. The pairs (p, q) with $p \sim P$ and $q \sim Q$ form a *box*. Its *exponent rectangle* is the set of the pairs (α, β) with $X^{\alpha} \in [P, 2P)$ and $X^{\beta} \in [Q, 2Q)$, and the distance of a box from a set of exponents is that of its exponent rectangle. In every sifting function $S(\mathcal{A}_u, \cdot)$ below, every prime factor of u satisfies the condition that defines the level, $|\varpi|^2 \geqslant z$ or $\varpi \succcurlyeq q$, so $S(\mathcal{A}_u, \cdot) = S'(\mathcal{A}_u, \cdot)$ by the definition (5) of S'. The sums and identities below are written for $\mathcal{A}$. Each sum is defined for $\mathcal{B}$ in the same way, and identities (29) and (31) hold with $\mathcal{B}$ in place of $\mathcal{A}$, since they follow from (6).

Lemma 9 gives the whole sum over a box an asymptotic formula, the treatment of Range 1 in Table 1. Its proof applies Buchstab's identity (6) at level μ^4, the linear estimate to the sum sifted at this level, and the bilinear estimate, with p as the variable u, to the sum that this identity subtracts.

Lemma 9 *Suppose that $W \leqslant P \leqslant X^{\zeta-\varepsilon}$, $Q \geqslant PX^{-2\theta}$ and $PQ \leqslant X^{\nu-\varepsilon}$. Then the sum of $S(\mathcal{A}_{pq}, q)$ over $p \sim P$ and $q \sim Q$ has an asymptotic formula.*

Proof. Since $Q \leqslant X^{\nu-\varepsilon}/W = X^{\rho-\varepsilon} < W \leqslant P$, the condition $q \prec p$ holds automatically. Buchstab's identity (6) at level μ^4 gives

$$\sum_{p\sim P,\, q\sim Q} S'(\mathcal{A}_{pq}, q) = \sum_{p\sim P,\, q\sim Q} S'(\mathcal{A}_{pq}, \mu^4) - \sum_{p\sim P,\, q\sim Q} \sum_{r\prec q,\ |r|^2\geqslant\mu^4} S'(\mathcal{A}_{pqr}, r).$$

The sum of $S'(\mathcal{A}_{pq}, \mu^4)$ is a configuration of type (C) without constraints. Its pieces fall under (P2) of Lemma 8 with $\mathcal{U} = \{p\}$, $\mathcal{V} = \{q\}$ and $\varepsilon/2$ in place of ε, since $P \leqslant X^{\zeta-\varepsilon}$, $Q \leqslant X^{\rho-\varepsilon}$ and $PQ \leqslant X^{\nu-\varepsilon}$.

In the sum of $S'(\mathcal{A}_{pqr}, r)$, only the primes r with $|r|^4 < X/(PQ)$ contribute. Indeed, for $|r|^4 \geqslant X/(PQ)$, a cofactor $t \neq 1$ has $|t|^2 \geqslant |r|^2$, so $|pqrt|^2 \geqslant PQ|r|^4 \geqslant X$. The cofactor $t = 1$ requires $|r|^2 \geqslant X/(8PQ)$. Since $PQ^2 \leqslant X^{\nu+\rho-2\varepsilon}$ and $\nu + \rho = (2 + 8\theta)/3 < 1$, this gives $|r|^2 > 2Q > |q|^2$, which contradicts $r \prec q$. Hence the sum of $S'(\mathcal{A}_{pqr}, r)$ is the configuration with variables p, q, r, the range $[\mu^4, (X/(PQ))^{1/2})$ for $|r|^2$, the constraint $r \prec q$, and the sieve condition of type (S) with i_s the index of r. Its requirement $r \preccurlyeq p$ holds, since $(X/(PQ))^{1/2} \leqslant X^{\tau}/P \leqslant X^{2\theta} < P$. All of its pieces, the window pieces included, fall under (P3) with $\mathcal{S} = \{p\}$ and $\varpi_{i_0} = r$. Indeed, $|p|^2 \geqslant W$, and the quotient $(X/2)/\sup|pr|^2$ of (P3) satisfies

$$\frac{X/2}{2P(X/(PQ))^{1/2}(1 + 1/H)} \geqslant \frac{(XQ/P)^{1/2}}{5} \geqslant \frac{W}{5},$$

since $Q \geqslant PX^{-2\theta}$. Lemma 7 completes the proof. □

In the proof of Lemma 11, Lemma 10 gives the sum over (p, q, r) in (29) an asymptotic formula. The cases (a) and (b) of (28) correspond to the factorizations $(pq) \cdot r$ and $p \cdot (qr)$ of Section 3.3.

Lemma 10 *Suppose that $Q \leqslant 2P$, $PQ \geqslant W$ and $X^{\tau}/(PQ) \geqslant X^{\varepsilon}$, and let $\mathcal{Q} \subset [Q, 2Q)$ and $\mathcal{I} \subset [X^{\tau}/(PQ), 2Q)$ be intervals with $\inf \mathcal{Q} \geqslant X^{\tau}/(PQ)$. Then*

$$\sum_{p\sim P,\ |q|^2\in\mathcal{Q},\ q\prec p} \sum_{|r|^2\in\mathcal{I},\ r\prec q} S\big(\mathcal{A}_{pqr}, X^{\tau}/(PQ)\big) \tag{27}$$

has an asymptotic formula if one of the following holds:

$$\text{(a) } PQ \leqslant X^{\zeta-\varepsilon},\ Q \leqslant X^{\rho-\varepsilon},\ PQ^2 \leqslant X^{\nu-\varepsilon}; \qquad \text{(b) } P \leqslant X^{\zeta-\varepsilon},\ Q^2 \leqslant X^{\rho-\varepsilon},\ PQ^2 \leqslant X^{\nu-\varepsilon}. \tag{28}$$

Proof. All of p, q, r have norm at least $X^{\tau}/(PQ) \geqslant X^{\varepsilon} \geqslant \mu^4$, so Buchstab's identity (6) gives

$$S\big(\mathcal{A}_{pqr}, X^{\tau}/(PQ)\big) = S'(\mathcal{A}_{pqr}, \mu^4) - \sum_{\mu^4\leqslant|s|^2<X^{\tau}/(PQ)} S'(\mathcal{A}_{pqrs}, s).$$

Summed as in (27), the term $S'(\mathcal{A}_{pqr}, \mu^4)$ is the configuration with variables p, q, r, ranges $[P, 2P)$, $\mathcal{Q}$, $\mathcal{I}$, constraints $q \prec p$ and $r \prec q$, and the sieve condition of type (C). The sum of $S'(\mathcal{A}_{pqrs}, s)$ is the configuration with the further variable s, of range $[\mu^4, X^{\tau}/(PQ))$, the same constraints, and the sieve condition of type (S) with i_s the index of s. Every tuple that satisfies the constraints has $s \prec r \prec q \prec p$, as type (S) requires. We apply Lemma 7 to both.

For the configuration of type (C), the pieces of hypothesis (a) of Lemma 7 fall under (P2), and the straddling pieces under (P1), with $\varepsilon/2$ in place of ε. There are no window pieces, since the window (24) is empty in type (C). In case (a) of (28), take $\mathcal{U} = \{p, q\}$ and $\mathcal{V} = \{r\}$: then $|pq|^2 < 4PQ$, $|r|^2 < 2Q$ and $|pqr|^2 < 8PQ^2$. In case (b), take $\mathcal{U} = \{p\}$ and $\mathcal{V} = \{q, r\}$: then $|p|^2 < 2P$, $|qr|^2 < 4Q^2$ and $|pqr|^2 < 8PQ^2$. In both cases (28) gives the conditions of (P1).

For the configuration of type (S), all pieces, the window pieces included, fall under (P3) with $\mathcal{S} = \{p, q\}$ and $\varpi_{i_0} = s$. Indeed, $|pq|^2 \geqslant PQ \geqslant W$. On every piece, $|s|^2 < (1 + 1/H)X^\tau/(PQ)$, so $(X/2)/\sup|pqs|^2 \geqslant W/9$. □

For the boxes with $PQ \geqslant W$, Lemma 11 applies Buchstab's identity (6) twice at the level $X^\tau/(PQ)$.

Lemma 11 *Suppose that $X^\varepsilon \leqslant Q \leqslant 2P$ and $W \leqslant PQ \leqslant X^{\tau-\varepsilon}$. Split*

$$\sum_{p\sim P,\, q\sim Q,\, q\prec p} S(\mathcal{A}_{pq}, q) = S_< + S_\geqslant$$

according as $|q|^2 < X^\tau/(PQ)$ or $|q|^2 \geqslant X^\tau/(PQ)$. Then $S_<$ has an asymptotic formula, and

$$S_\geqslant = \sum_{p,q} S\big(\mathcal{A}_{pq}, X^\tau/(PQ)\big) - \sum_{p,q} \sum_{\substack{r\prec q\\ |r|^2\geqslant X^\tau/(PQ)}} S\big(\mathcal{A}_{pqr}, X^\tau/(PQ)\big) + \sum_{p,q} \sum_{\substack{s\prec r\prec q\\ |s|^2\geqslant X^\tau/(PQ)}} S(\mathcal{A}_{pqrs}, s), \quad (29)$$

where each $\sum_{p,q}$ runs over $p \sim P$, $q \sim Q$, $q \prec p$, $|q|^2 \geqslant X^\tau/(PQ)$. If moreover (a) or (b) of (28) holds, then the sums in (29) over (p, q) and over (p, q, r) have asymptotic formulas.

Proof. Lemma 7 treats $S_<$, Lemma 4 the sum over (p, q) in (29), and Lemma 10 the sum over (p, q, r). For $n = pq$ with $p \sim P$ and $q \sim Q$,

$$W \leqslant |n|^2 < 4PQ \leqslant X^{\tau-\varepsilon/2}, \qquad \frac{X^\tau}{PQ} < \frac{4X^\tau}{|n|^2}. \quad (30)$$

The sum $S_<$ is the configuration with variables p, q, ranges $[P, 2P)$ and $[Q, \min(2Q, X^\tau/(PQ)))$, the constraint $q \prec p$, and the sieve condition of type (S) with i_s the index of q. In Lemma 7, its pieces of hypothesis (a) have, by (23), the level $\xi_{\lambda_q} \leqslant |q|^2 < X^\tau/(PQ)$, and they fall under (25) of Lemma 8 with $n = pq$ and $\varepsilon/2$ in place of ε, by (30). Its straddling pieces, with p and q in the same short interval, fall under (P1) with $\varepsilon/2$ in place of ε, by (30). Its window pieces, with ϖ' in the short interval of q by (24), fall under (P3) with $\mathcal{S} = \{p, q\}$ and $\varpi_{i_0} = \varpi'$, since $|pq|^2 \geqslant W$ and $|pq\varpi'|^2 < 4(1 + 1/H)X^\tau$, so that $(X/2)/\sup|pq\varpi'|^2 \geqslant W/9$.

For $|q|^2 \geqslant X^\tau/(PQ)$, Buchstab's identity (6), applied first to $S(\mathcal{A}_{pq}, q)$ and then to each $S(\mathcal{A}_{pqr}, r)$ with $|r|^2 \geqslant X^\tau/(PQ)$, gives (29). The sum over (p, q) in (29) is $\sum_{n\neq 0} a_n S(\mathcal{A}_n, X^\tau/(PQ))$, with $n = pq$ and a_n the number of pairs (p, q) in the range of $\sum_{p,q}$ with $pq = n$. Lemma 4 applies to it with $\varepsilon/2$ in place of ε, by (30). The sum over (p, q, r) is the case $\mathcal{Q} = [\max(Q, X^\tau/(PQ)), 2Q)$ and $\mathcal{I} = [X^\tau/(PQ), 2Q)$ of Lemma 10, whose hypothesis $X^\tau/(PQ) \geqslant X^\varepsilon$ holds because $PQ \leqslant X^{\tau-\varepsilon}$. □

For the boxes with $PQ < W$, the level $X^\tau/(PQ)$ of (29) exceeds $z(PQ)$, the level that Lemma 5 admits up to a constant factor, and Lemma 12 applies Buchstab's identity (6) at levels given by z instead.

Lemma 12 *Let $j \in \{0, 1\}$ and $0 < 4\varepsilon < 5\theta/2 - 1/4$. Suppose that $X^\varepsilon \leqslant Q \leqslant 2P$ and $WX^{-(j+1)\theta} \leqslant PQ < WX^{-j\theta}$, so that $j(PQ) = j$ and $z(PQ) = (X^\tau/(PQ))^{1/(j+2)}$. Split*

$$\sum_{p\sim P,\, q\sim Q,\, q\prec p} S(\mathcal{A}_{pq}, q) = S_< + S_\geqslant$$

according as $|q|^2 < z(PQ)$ or $|q|^2 \geqslant z(PQ)$. Then $S_<$ has an asymptotic formula, and

$$\begin{aligned} S_\geqslant = \sum_{p,q} S\big(\mathcal{A}_{pq}, z(PQ)\big) - \sum_{p,q} \sum_{\substack{r\prec q\\ |r|^2\geqslant z(PQ)}} S\Big(\mathcal{A}_{pqr}, \Big(\frac{X^\tau}{PQ|r|^2}\Big)^{1/(j+1)}\Big) \\ + \sum_{p,q} \sum_{\substack{r\prec q\\ |r|^2\geqslant z(PQ)}} \sum_{\substack{s\prec r\\ |s|^{2(j+1)}\geqslant X^\tau/(PQ|r|^2)}} S(\mathcal{A}_{pqrs}, s), \end{aligned} \quad (31)$$

where each $\sum_{p,q}$ runs over $p \sim P$, $q \sim Q$, $q \prec p$, $|q|^2 \geqslant z(PQ)$. The sums in (31) over (p, q) and over (p, q, r) have asymptotic formulas.

Proof. We first derive (31) and prove the bounds (33) to (36). With these bounds, Lemmas 4 and 5 give the sum over (p,q) in (31) an asymptotic formula. Lemma 7 gives $S_<$ and the sum over (p,q,r) asymptotic formulas: the pieces of its hypothesis (a) fall under (P4) of Lemma 8, and the straddling and window pieces under (P1) or (P3).

For $|q|^2 \geqslant z(PQ)$, Buchstab's identity (6) gives

$$S(\mathcal{A}_{pq},q) = S\big(\mathcal{A}_{pq},z(PQ)\big) - \sum_{\substack{r\prec q\\ |r|^2\geqslant z(PQ)}} S(\mathcal{A}_{pqr},r).$$

For such r, $|r|^{2(j+2)} \geqslant z(PQ)^{j+2} = X^\tau/(PQ)$, so the level of the sum over (p,q,r) in (31), which serves as the function m_s of a sieve condition of type (M) of Section 5, satisfies

$$m_s(p,q,r) := \Big(\frac{X^\tau}{PQ|r|^2}\Big)^{1/(j+1)} \leqslant |r|^2. \tag{32}$$

So (6) applies again to $S(\mathcal{A}_{pqr},r)$, at the level $m_s(p,q,r)$, and (31) follows.

For a prime q' with $|q'|^2 \leqslant 4Q$, which below is r or the prime ϖ' of a window piece, the bounds $Q^2 \leqslant 2PQ$ and $PQ < W$ give $|pqq'|^2 < 16PQ^2 < 32(PQ)^{3/2} < 32X^{3\sigma/2}$. Since $3\sigma/2 = \tau - (5\theta/2 - 1/4)$ and $4\varepsilon < 5\theta/2 - 1/4$,

$$|pq|^2|q'|^2 \leqslant X^{\tau-3\varepsilon} \qquad (|q'|^2 \leqslant 4Q). \tag{33}$$

If $|r|^2 \geqslant z(PQ)$, then $|pqr|^2 \geqslant PQ\,z(PQ) = (PQ)^{(j+1)/(j+2)}X^{\tau/(j+2)}$, and $PQ \geqslant WX^{-(j+1)\theta}$ gives

$$|pqr|^2 \geqslant WX^{\theta/2} \quad (j=0), \qquad |pqr|^2 \geqslant WX^{-2\theta/3} \quad (j=1). \tag{34}$$

For $|r|^2 < 2Q$, the level (32) satisfies

$$m_s(p,q,r) \geqslant \Big(\frac{X^\tau}{8PQ^2}\Big)^{1/(j+1)} \geqslant \frac{X^{(\tau-3\sigma/2)/2}}{8} \geqslant X^{2\varepsilon} \geqslant \mu^4. \tag{35}$$

Let $n = pq$ with $p \sim P$ and $q \sim Q$. Then $|n|^2 \geqslant PQ \geqslant WX^{-(j+1)\theta}$, so $j(|n|^2) \leqslant j$ if $|n|^2 < W$. Also $|n|^2 < 4PQ < 4WX^{-j\theta}$, so $|n|^2 \geqslant W$ only for $j=0$, and then $|n|^2 < 4W \leqslant X^{\tau-\varepsilon}$. Hence

$$\begin{aligned} z(|n|^2) &\geqslant \Big(\frac{X^\tau}{4PQ}\Big)^{1/(j+2)} > \frac{z(PQ)}{4} \quad (|n|^2 < W),\\ z(PQ) &= \Big(\frac{X^\tau}{PQ}\Big)^{1/2} \leqslant \frac{X^\tau}{4PQ} \leqslant \frac{X^\tau}{|n|^2} \quad (|n|^2 \geqslant W). \end{aligned} \tag{36}$$

The sum over (p,q) in (31) is $\sum_{n\neq 0} a_n S(\mathcal{A}_n, z(PQ))$, with $n = pq$ and a_n the number of pairs (p,q) in the range of $\sum_{p,q}$ with $pq = n$. By (36), Lemma 5 applies to the n with $|n|^2 < W$, and Lemma 4 to the others, both with the constant level $z(PQ)$.

The sum $S_<$ is the configuration with variables p,q, ranges $[P,2P)$ and $[Q,\min(2Q,z(PQ)))$, the constraint $q \prec p$, and the sieve condition of type (S) with i_s the index of q. In Lemma 7, its pieces of hypothesis (a) have, by (23), the level $y = \xi_{\lambda_q} \leqslant |q|^2 < z(PQ)$, and they fall under (P4) with $n = pq$ by (36). Its straddling pieces and window pieces fall under (P1) by (33).

The sum over (p,q,r) in (31) is the configuration with variables p,q,r, constraints $q \prec p$ and $r \prec q$, the ranges $[P,2P)$, $[\max(Q,z(PQ)),2Q)$ and $[z(PQ),2Q)$, and the sieve condition of type (M) with the level $m_s(p,q,r)$ of (32). Its requirement $\mu^4 \leqslant m_s(p,q,r) \leqslant |r|^2$ holds by (32) and (35). In Lemma 7, a piece of hypothesis (a) has, by (23), the level $y = \inf m_s$ over the piece. Every product $n = pqr$ of the piece has $|n|^2 < 4PQ|r|^2$, so

$$y \leqslant m_s(p,q,r) < \Big(\frac{4X^\tau}{|n|^2}\Big)^{1/(j+1)}.$$

For $j=0$, every n has $W \leqslant |n|^2 \leqslant X^{\tau-3\varepsilon}$ by (34) and (33), so the piece falls under (25). For $j=1$, an n with $|n|^2 \geqslant W$ has $|n|^2 \leqslant X^{\tau-3\varepsilon}$ by (33) and $y \leqslant y^2 < 4X^\tau/|n|^2$, which is (25). An n with $|n|^2 < W$ has $|n|^2 \geqslant WX^{-2\theta/3}$ by (34), so $j(|n|^2) = 0$ and $y < 2(X^\tau/|n|^2)^{1/2} = 2z(|n|^2)$, which is (26).

The straddling pieces fall under (P1) by (33). A window piece adds a prime ϖ' with $|\varpi'|^2$ within the factor $(1+1/H)^2$ of $m_s(p,q,r)$, by (24). For $j=1$ it falls under (P1), because $|pqr\varpi'|^2 \leqslant 8(X^\tau|pqr|^2)^{1/2} \leqslant X^{\tau-\varepsilon}$ by (33). For $j=0$ it falls under (P3) with $\mathcal{S}=\{p,q,r\}$ and $\varpi_{i_0}=\varpi'$. Indeed, $|pqr|^2 \geqslant W$ by (34), and $|pqr\varpi'|^2 \leqslant 5X^\tau$ on the piece, so $(X/2)/\sup|pqr\varpi'|^2 \geqslant W/10$. □

7 The ranges and the discarded regions

This section proves identity (12), defines the regions of Table 1, and checks the treatment of each range.

Lemma 13 *Let Π_1, Π_2 and $\sum^*$ be as in Section 3. Identity (12) holds for $\mathcal{E}\in\{\mathcal{A},\mathcal{B}\}$. The terms on the right of (12) other than $\sum^*$ have asymptotic formulas. A prime p with $|p|^2=X^\alpha$ lies in Π_2 exactly when $\theta\leqslant\alpha<\tau/5$ or $\sigma-2\theta\leqslant\alpha<\tau/4$. The interval $[\sigma-2\theta,\tau/4)$ is empty for $\theta\leqslant 3/26$.*

Proof. Buchstab's identity (6) gives

$$S(\mathcal{E},X^{1/2}) = S(\mathcal{E},X^\theta) - \sum_{X^\theta\leqslant|p|^2<X^{1/2}} S(\mathcal{E}_p,p),$$

and $S(\mathcal{E}_p,p)=S'(\mathcal{E}_p,p)$. For $p\in\Pi_1$, the identity of (6) for $S'(\mathcal{E}_u,q)$, with $u=q=p$ and $w=z(|p|^2)\leqslant|p|^2$, gives

$$S'(\mathcal{E}_p,p) = S\big(\mathcal{E}_p,z(|p|^2)\big) - \sum_{\substack{q\prec p\\ |q|^2\geqslant z(|p|^2)}} S(\mathcal{E}_{pq},q).$$

Together they give (12).

The term $S(\mathcal{E},X^\theta)$ of (12) has an asymptotic formula by Corollary 6. In the sum of $S(\mathcal{E}_p,z(|p|^2))$ over $p\in\Pi_1$, the level $z(|p|^2)$ does not exceed $|p|^2$, so $S=S'$. For $|p|^2\geqslant W$ the level is $X^\tau/|p|^2$ and $|p|^2<X^{1/2}=X^{\tau-\theta}$, so Lemma 4 applies. For $|p|^2<W$, Lemma 5 applies with the level $z(|p|^2)$.

In the sum over $p\in\Pi_2$, the term $S(\mathcal{E}_p,p)$ differs from $S'(\mathcal{E}_p,|p|^2)$ only by the cofactors divisible by the other prime p' of norm $|p|^2$, when it exists and $p'\prec p$. So $0\leqslant S'(\mathcal{E}_p,|p|^2)-S(\mathcal{E}_p,p)\leqslant|\mathcal{E}_{pp'}|$. For $p\in\Pi_2$, $|p|^2<z(|p|^2)$ and $|p|^2<W$, so Lemma 5 gives the sum of $S'(\mathcal{E}_p,|p|^2)$ over $p\in\Pi_2$ an asymptotic formula. Since $|pp'|^2=|p|^4<X^{\tau/2}$, Lemmas 1(a) and 18(a) give

$$\sum_{p\in\Pi_2}|\mathcal{A}_{pp'}| = \frac{2}{\pi W}\sum_{p\in\Pi_2}|\mathcal{B}_{pp'}| + \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big), \qquad \sum_{p\in\Pi_2}|\mathcal{B}_{pp'}| \ll X\sum_{|p|^2\geqslant X^\theta}|p|^{-4} \ll X^{1-\theta}.$$

So the sums of $|\mathcal{A}_{pp'}|$ and of $2|\mathcal{B}_{pp'}|/(\pi W)$ over $p\in\Pi_2$ are $\mathcal{O}(XW^{-1}(\log X)^{-A})$, and the sum of $S(\mathcal{E}_p,p)$ over $p\in\Pi_2$ has an asymptotic formula.

By the definition of β_0, a prime p with $|p|^2=X^\alpha$ and $\theta\leqslant\alpha<1/2$ lies in Π_2 exactly when $\beta_0(\alpha)>\alpha$. For $\alpha\geqslant\sigma$, $\beta_0(\alpha)=\tau-\alpha\leqslant\alpha$. For $\alpha<\sigma$ and $j=j(X^\alpha)$, the inequality $\beta_0(\alpha)>\alpha$ means $\alpha<\tau/(j+3)$. For $j=0$ this is impossible, because $j=0$ gives $\alpha\geqslant\sigma-\theta>\tau/3$. For $j=1$ it gives $\alpha\in[\sigma-2\theta,\tau/4)$, and $\tau/4\leqslant\sigma-2\theta$ for $\theta\leqslant 3/26$. For $j=2$ it gives $\theta\leqslant\alpha<\tau/5$, since $\theta>\sigma-3\theta$ and $\tau/5<\sigma-2\theta$. For $j\geqslant 3$, $\alpha<\sigma-3\theta<\theta$, which is excluded. □

At $\theta=0.1129$ the primes of Π_2 have $\alpha\in[0.1129,0.12258)$, so D begins at $\alpha=\tau/5=0.12258$ in Figure 1.

In the sum $\sum^*$, a term with $|p|^2|q|^4\geqslant X$ has cofactor $t=1$, because a cofactor $t\neq 1$ has $|t|^2\geqslant|q|^2$ and would give $|pqt|^2\geqslant|p|^2|q|^4\geqslant X$. Then $pq\in\mathcal{E}$ and $q\prec p$ force $|p|^4\geqslant X/2$ and $|q|^2\geqslant X^{1/2}/2$, and the number of such pairs with $pq\in\mathcal{B}$ is $\mathcal{O}(X/(\log X)^2)$. These terms are non-negative and enter (12) with the sign +, so we discard them. Their contribution tends to 0 after division by $S(\mathcal{B},X^{1/2})$, which is of order $X/\log X$ by (46). The remaining pairs have exponents in the domain D of (13).

On each part of Ranges 1 to 6, the hypotheses of the lemmas used there are linear inequalities in α, β and θ. Appendix D lists the inequalities between the parameters that these hypotheses use, with their values at $\theta=0.1129$. On each range, the discarded region consists of the pairs on which no lemma of Section 6 applies.

Range 1 consists of the pairs with $\sigma \leqslant \alpha < 1/2$. Here $\beta_0(\alpha) = \tau - \alpha$ and $\beta < (1-\alpha)/2 < \alpha$. Lemma 9 gives an asymptotic formula where $\alpha \leqslant \zeta$, $\beta \geqslant \alpha - 2\theta$ and $\alpha + \beta \leqslant \nu$. The discarded region is

$$\mathsf{D}_1 := \big\{\sigma \leqslant \alpha < 1/2,\ \tau - \alpha \leqslant \beta < (1-\alpha)/2\big\} \setminus \big\{\alpha - 2\theta \leqslant \beta \leqslant \nu - \alpha\big\}. \tag{37}$$

The set subtracted in D_1 is empty for $\alpha > \chi$, where $\chi := (\nu + 2\theta)/2$. Since $\chi < \zeta$, the condition $\alpha \leqslant \zeta$ of Lemma 9 holds wherever this set is non-empty. The exponent χ is that of the hypothesis $P \ll X^\chi$ of Lemma 11 of Harman and Lewis [10] (Appendix E).

Range 2 consists of the pairs with $1/3 \leqslant \alpha < \sigma$. Here $j(X^\alpha) = 0$, so $\beta_0(\alpha) = (\tau-\alpha)/2$, and $\beta < (1-\alpha)/2$. Lemma 11 applies in case (a) of (28) where $\alpha+\beta \leqslant \zeta$, since there $\beta \leqslant \zeta - 1/3 < \rho$ and $\alpha + 2\beta \leqslant 2\zeta - 1/3 < \nu$. It applies in case (b) on the region R of (15), since there $\alpha < \sigma < \zeta$, $2\beta \leqslant \rho$, $\alpha + 2\beta < \sigma + \rho = \nu$ and $\alpha + \beta < \sigma + \rho/2 < \tau$. In both cases $\alpha + \beta \geqslant (\alpha + \tau)/2 > \sigma$ and $\beta \geqslant (\tau - \alpha)/2$, so $PQ \geqslant W$ and $S_< = 0$. Where Lemma 11 applies, only the third sum of (29) is discarded. The two-dimensional discarded region is

$$\mathsf{D}_2 := \big\{1/3 \leqslant \alpha < \sigma,\ \max(\zeta - \alpha, (\tau-\alpha)/2) \leqslant \beta < (1-\alpha)/2,\ \beta > \rho/2\big\}. \tag{38}$$

Range 3 consists of the pairs with $\rho \leqslant \alpha < 1/3$. Here $\beta_0(\alpha) = (\tau - \alpha)/2$ and $\beta < \alpha$. Lemma 11 applies in case (a) where $\alpha + \beta \leqslant \zeta$, since there $\beta \leqslant \zeta - \rho < \rho$, $\alpha + 2\beta \leqslant 2\zeta - \rho = \nu$ and $\alpha + \beta \geqslant (\rho + \tau)/2 > \sigma$. The discarded region is

$$\mathsf{D}_4 := \big\{\rho \leqslant \alpha < 1/3,\ \zeta - \alpha \leqslant \beta < \alpha\big\}. \tag{39}$$

Range 4 consists of the pairs with $\sigma - \theta \leqslant \alpha < \rho$. Here $\beta_0(\alpha) = (\tau - \alpha)/2$. Lemma 11 applies in case (a) where $\alpha + 2\beta \leqslant \nu$, since there $\alpha + \beta \leqslant (\alpha + \nu)/2 < (\rho + \nu)/2 = \zeta$, $\beta \leqslant (\nu - \sigma + \theta)/2 = (\rho + \theta)/2 < \rho$ and $\alpha + \beta \geqslant (\alpha + \tau)/2 \geqslant (1 - \theta)/2 > \sigma$. The discarded region is

$$\mathsf{D}_6 := \big\{\sigma - \theta \leqslant \alpha < \rho,\ (\nu - \alpha)/2 \leqslant \beta < \alpha\big\}. \tag{40}$$

Range 5 consists of the pairs with $\alpha < \sigma - \theta$ and $\alpha + \beta \geqslant \sigma - \theta$. Here $j(X^\alpha) \geqslant 1$, so $\beta_0(\alpha) \leqslant (\tau - \alpha)/3$, and pairs with $\beta < (\tau - \alpha)/2$ occur. Where $\alpha + \beta \geqslant \sigma$ and $\beta < (\tau - \alpha)/2$, the sum equals $S_<$ of Lemma 11 and has an asymptotic formula. Where $\alpha + \beta \geqslant \sigma$, $\beta \geqslant (\tau - \alpha)/2$ and $\alpha + 2\beta \leqslant \nu$, Lemma 11 applies in case (a), since there $\beta < \alpha < \sigma - \theta < \rho$ and $\alpha + \beta \leqslant (\alpha + \nu)/2 < (\sigma - \theta + \nu)/2 < \zeta$. Where $\alpha + \beta < \sigma$, $j(X^{\alpha+\beta}) = 0$, and Lemma 12 with $j = 0$ applies. The discarded region is

$$\mathsf{D}_8 := \big\{\nu/3 \leqslant \alpha < \sigma - \theta,\ (\nu - \alpha)/2 \leqslant \beta < \alpha\big\}. \tag{41}$$

Range 6 consists of the pairs with $\alpha + \beta < \sigma - \theta$. Since $p \in \Pi_1$, Lemma 13 gives $\alpha \geqslant \tau/5 > \sigma - 3\theta$, so $j(X^\alpha) \leqslant 2$ and $\beta \geqslant \beta_0(\alpha) \geqslant (\tau - \alpha)/4$. Hence $\alpha + \beta \geqslant (3\alpha + \tau)/4 \geqslant 2\tau/5 > \sigma - 2\theta$, so $j(X^{\alpha+\beta}) = 1$, and Lemma 12 with $j = 1$ applies. Since $\beta \geqslant (\tau - \alpha)/4$ is equivalent to $|q|^2 \geqslant z(|pq|^2)$, $S_< = 0$ on every box at distance at least ε from the boundary of D.

In the third sum of (29), the exponents satisfy $\tau - \alpha - \beta \leqslant \delta < \gamma < \beta$. In that of (31), they satisfy

$$\frac{\tau - \alpha - \beta}{j+2} \leqslant \gamma < \beta, \qquad \frac{\tau - \alpha - \beta - \gamma}{j+1} \leqslant \delta < \gamma.$$

With these limits, the third sums are discarded over the following regions of $(\alpha, \beta, \gamma, \delta)$:

$$\begin{aligned}
&\mathsf{E}_3 : 1/3 \leqslant \alpha < \sigma,\ (\tau - \alpha)/2 \leqslant \beta \leqslant \zeta - \alpha,\ \tau - \alpha - \beta \leqslant \delta < \gamma < \beta;\\
&\mathsf{E}_\mathsf{R} : (\alpha, \beta) \in \mathsf{R},\ \tau - \alpha - \beta \leqslant \delta < \gamma < \beta;\\
&\mathsf{E}_5 : \rho \leqslant \alpha < 1/3,\ (\tau - \alpha)/2 \leqslant \beta \leqslant \zeta - \alpha,\ \tau - \alpha - \beta \leqslant \delta < \gamma < \beta;\\
&\mathsf{E}_7 : \sigma - \theta \leqslant \alpha < \rho,\ (\tau - \alpha)/2 \leqslant \beta \leqslant (\nu - \alpha)/2,\ \tau - \alpha - \beta \leqslant \delta < \gamma < \beta;\\
&\mathsf{E}_9 : \alpha < \sigma - \theta,\ (\tau - \alpha)/2 \leqslant \beta \leqslant \min((\nu - \alpha)/2, \alpha),\ \tau - \alpha - \beta \leqslant \delta < \gamma < \beta;\\
&\mathsf{E}_{10} : \tau/4 \leqslant \alpha \leqslant \sigma - \theta,\ (\tau - \alpha)/3 \leqslant \beta < \min(\alpha, \sigma - \alpha),\\
&\qquad (\tau - \alpha - \beta)/2 \leqslant \gamma < \beta,\ \tau - \alpha - \beta - \gamma \leqslant \delta < \gamma;\\
&\mathsf{E}_{11} : \tau/5 \leqslant \alpha \leqslant \sigma - 2\theta,\ (\tau - \alpha)/4 \leqslant \beta < \min(\alpha, \sigma - \theta - \alpha),\\
&\qquad (\tau - \alpha - \beta)/3 \leqslant \gamma < \beta,\ (\tau - \alpha - \beta - \gamma)/2 \leqslant \delta < \gamma.
\end{aligned} \tag{42}$$

In E_{10}, $\alpha \geqslant \tau/4$ and $\beta \geqslant (\tau-\alpha)/3$ give $\alpha+\beta \geqslant (2\alpha+\tau)/3 \geqslant \tau/2 > \sigma-\theta$, so the pairs of E_{10} lie in Range 5. The region E_{11} contains the set of exponents over which the third sum is discarded in Range 6, since $\beta_0(\alpha) \geqslant (\tau-\alpha)/4$. For $\theta < 3/26$, the primes with $\alpha \in [\tau/4, \sigma-2\theta)$ have $j(X^\alpha) = 2$ and $\beta_0(\alpha) = (\tau-\alpha)/4$. Their pairs lie in Ranges 5 and 6, where Lemma 12 does not depend on $j(X^\alpha)$, and the discarded third sums of these pairs lie in E_{10} and E_{11}.

Lemmas 14 and 16 need a positive lower bound for the exponents, and Lemma 16 an upper bound for their sum. All exponents on D and on the regions (42) are at least

$$c(\theta) := \min\Big(4\theta^2,\ \frac{5\theta}{2}-\frac{1}{4},\ \frac{26\theta-1}{24},\ \frac{1-2\theta}{6}\Big) > 0, \tag{43}$$

and their sums satisfy

$$\alpha+\beta < 1-c(\theta) \quad \text{on } \mathsf{D}, \qquad \alpha+\beta+\gamma+\delta < 1-c(\theta) \quad \text{on the regions (42)}. \tag{44}$$

On D, $\alpha \geqslant \theta$ and $\beta \geqslant \beta_0(\alpha)$. For $\alpha \geqslant \sigma$, $\beta_0(\alpha) = \tau-\alpha > \theta$. For $\alpha < \sigma$ and $j = j(X^\alpha)$, $\alpha \geqslant \theta$ gives $j+2 < (\sigma+\theta)/\theta = 1/(2\theta)$, so

$$\beta_0(\alpha) = \frac{\tau-\alpha}{j+2} > \frac{2\theta}{j+2} > 4\theta^2.$$

Also $\alpha+\beta < 1-\beta \leqslant 1-c(\theta)$ on D, since $\beta < (1-\alpha)/2$.

On the regions (42), $\alpha \geqslant \beta \geqslant \gamma \geqslant \delta$, so it suffices to bound δ from below and $\alpha+3\beta$ from above. On E_3, E_R, E_5, E_7 and E_9, the lower limit of δ is $\tau-\alpha-\beta$, so an upper bound for $\alpha+\beta$ bounds δ from below. On E_3 and E_5, $\alpha+\beta \leqslant \zeta$. On E_7, $\alpha+\beta \leqslant (\alpha+\nu)/2 < (\rho+\nu)/2 = \zeta$. On E_9, $\alpha+\beta \leqslant (\alpha+\nu)/2 < (\sigma-\theta+\nu)/2 < \zeta$, by Table 4. On E_R, $\alpha+\beta < \sigma+\rho/2$. Hence, on these regions,

$$\delta \geqslant \tau-\alpha-\beta \geqslant \min\Big(\tau-\zeta,\ \tau-\sigma-\frac{\rho}{2}\Big) = \min\Big(\frac{1-2\theta}{6},\ \frac{26\theta-1}{24}\Big).$$

On E_{10}, $\alpha+\beta < \sigma$ and $\gamma < \beta < \sigma/2$, so $\delta \geqslant \tau-\alpha-\beta-\gamma > \tau-3\sigma/2 = 5\theta/2-1/4$. On E_{11}, $\alpha+\beta < \sigma-\theta$ and $\gamma < \beta < (\sigma-\theta)/2$, so $\delta > 2\theta-1/8 \geqslant 5\theta/2-1/4$.

For the sums of the exponents in (44), $\alpha+3\beta$ is at most $3\zeta-2/3$ on E_3, $\sigma+3\rho/2$ on E_R, $3\zeta-2\rho$ on E_5, $(3\nu-\sigma+\theta)/2$ on E_7, $4\nu/3$ on E_9, 2σ on E_{10} and $2\sigma-2\theta$ on E_{11}. Each of these bounds is below $15/16$ for $\theta < 1/8$, and $15/16 < 1-c(\theta)$ since $c(\theta) \leqslant 4\theta^2 < 1/16$.

8 The four-dimensional sums

This section proves that the third sums of (29) and (31) have asymptotic formulas on the regions G and L of (16). Let $p \sim P$, $q \sim Q$ be a box on which Lemma 11 or Lemma 12 gives identity (29) or (31). For powers R and R' of 2, the quadruples (p,q,r,s) of the third sum of that identity with $r \sim R$ and $s \sim R'$ form a *four-dimensional box*. Its *exponent box* is the set of the $(\alpha,\beta,\gamma,\delta)$ with $X^\alpha \in [P,2P)$, $X^\beta \in [Q,2Q)$, $X^\gamma \in [R,2R)$ and $X^\delta \in [R',2R')$. Let G_ε and L_ε be the regions defined by the inequalities of (16) with margin ε.

Lemma 14 *If the exponent box of a four-dimensional box lies in G_ε or in L_ε, then the sum of $S(\mathcal{A}_{pqrs}, s)$ over the box has an asymptotic formula.*

Proof. By (43), all exponents on the regions (42) are at least $c(\theta)$, so all ranges lie in $[\mu^4, X]$. The sum is the configuration with variables p,q,r,s, the box ranges, the constraints $q \prec p$, $r \prec q$ and $s \prec r$, the lower limits of $|r|^2$ and $|s|^2$ in (29) or (31), and the sieve condition of type (S) with i_s the index of s. The lower limits are constants for (29). For (31), the lower limit of s is the monomial constraint $|r|^2|s|^{2(j+1)} \geqslant X^\tau/(PQ)$. By Lemma 7, it suffices to check the pieces.

Suppose the exponent box lies in G_ε. At one point of the exponent box, a non-empty set $\mathcal{S} \subset \{p,q,r\}$ satisfies $\sigma \leqslant x_\mathcal{S} \leqslant \tau-\delta$ with margin ε. The exponent box has diameter $\mathcal{O}(1/\log X)$, so $\mathcal{S}$ satisfies these inequalities with margin $\varepsilon/2$ on the whole exponent box. Every piece of hypotheses (a), (b) and (c) of Lemma 7 falls under (P3) of Lemma 8 with this $\mathcal{S}$ and $\varpi_{i_0} = s$. Indeed, the primes of $\mathcal{S}$ have product of norm at least $X^{\sigma+\varepsilon/4}$, and $(X/2)/\sup|s\prod_{i\in\mathcal{S}}\varpi_i|^2$ is at least $X^{1-x_\mathcal{S}-\delta-\varepsilon/4} \geqslant X^{\sigma+\varepsilon/4}$.

Suppose the exponent box lies in L_ε. Buchstab's identity (6) at level μ^4, with s' running over primes, gives

$$S'(\mathcal{A}_{pqrs}, s) = S'(\mathcal{A}_{pqrs}, \mu^4) - \sum_{s' \prec s,\ |s'|^2 \geqslant \mu^4} S'(\mathcal{A}_{pqrss'}, s'). \tag{45}$$

Summed over the box, the term $S'(\mathcal{A}_{pqrs}, \mu^4)$ of (45) is the configuration with the same variables and constraints, and with the sieve condition of type (C). As for G, one of the alternatives that define I in (16) holds with margin $\varepsilon/2$ on the whole exponent box, so the pieces of hypothesis (a) fall under (P2), and the straddling pieces under (P1), with $\varepsilon/4$ in place of ε. The sum of $S'(\mathcal{A}_{pqrss'}, s')$ is the configuration with the further prime variable s', the constraint $s' \prec s$, and the sieve condition of type (S) with i_s the index of s'. As for G, a non-empty set $\mathcal{S} \subset \{p, q, r, s\}$ satisfies $\sigma \leqslant x_{\mathcal{S}} \leqslant \tau - \delta$ with margin $\varepsilon/2$ on the whole exponent box. Since $|s'|^2 \leqslant |s|^2 < 2X^\delta$, the quotient $(X/2)/\sup|s' \prod_{i\in\mathcal{S}} \varpi_i|^2$ is at least $X^{1-x_{\mathcal{S}}-\delta-\varepsilon/4} \geqslant X^{\sigma+\varepsilon/4}$. So every piece, the window pieces included, falls under (P3) with this $\mathcal{S}$ and $\varpi_{i_0} = s'$. Both terms of (45) have asymptotic formulas. □

9 The discarded sums and the lower bound

This section proves Lemma 16, the lower bound announced in Section 3. An element of $\mathcal{B}$ without prime factors of norm below $X^{1/2}$ is a Gaussian prime. So, with $\pi_G(y)$ the number of Gaussian primes of norm less than y, Lemma 17(d) gives

$$S(\mathcal{B}, X^{1/2}) = \pi_G(X) - \pi_G(X/2) = (1 + o(1))\frac{X}{2\log X}. \tag{46}$$

The next lemma evaluates the discarded sums over $\mathcal{B}$.

Lemma 15 *Let $K \geqslant 1$ and $c_1 > 0$, and let $\mathcal{R} \subset \mathbb{R}^K$ be a bounded set whose boundary has Lebesgue measure 0, on which $y_i \geqslant c_1$ and $y_K \leqslant y_i$ for all i, and $\sum_{i=1}^K y_i \leqslant 1 - c_1$. With $\boldsymbol{y}(\boldsymbol{\varpi}) := (\log|\varpi_i|^2/\log X)_{i\leqslant K}$ for a tuple of primes $\boldsymbol{\varpi} = (\varpi_1, \dots, \varpi_K)$,*

$$\sum_{\substack{\boldsymbol{y}(\boldsymbol{\varpi})\in\mathcal{R} \\ \varpi_K \prec \varpi_i \ (i<K)}} S(\mathcal{B}_{\varpi_1\cdots\varpi_K}, \varpi_K) = S(\mathcal{B}, X^{1/2})\Big(\int_{\mathcal{R}} \omega\Big(\frac{1-\sum_{i=1}^K y_i}{y_K}\Big)\frac{d\boldsymbol{y}}{y_1\cdots y_{K-1}y_K^2} + o(1)\Big). \tag{47}$$

Appendix C proves Lemma 15. The lower bound below uses (47) only as an upper bound for discarded sums.

Lemma 16 *For every small $\varepsilon > 0$ there is a number $\Phi_\varepsilon(\theta)$, with $\Phi_\varepsilon(\theta) \to \Phi(\theta)$ as $\varepsilon \to 0$, such that*

$$S(\mathcal{A}, X^{1/2}) \geqslant \frac{2}{\pi W} S(\mathcal{B}, X^{1/2})\big(1 - \Phi_\varepsilon(\theta) - o(1)\big). \tag{48}$$

The same holds with Φ_{P} in place of Φ. In particular, if $\Phi(\theta) < 1$ or $\Phi_{\mathsf{P}}(\theta) < 1$, then $S(\mathcal{A}, X^{1/2}) \gg XW^{-1}/\log X$.

Proof. Identity (12) holds for $\mathcal{A}$ and for $\mathcal{B}$, and its terms other than $\sum^*$ have asymptotic formulas by Lemma 13. The domain D of (13) is divided by Ranges 1 to 6 into finitely many parts, which are polytopes, and Section 7 gives the treatment of each part. The hypotheses of the lemma applied on a part are finitely many linear inequalities in (α, β) that hold on the closed part (Section 7 and Appendix D). Hence they hold with margin proportional to ε at the points of the part at distance at least ε from its boundary.

Split $\sum^*$ into the $\mathcal{O}((\log X)^2)$ dyadic boxes $p \sim P$, $q \sim Q$, whose exponent rectangles have side $\log 2/\log X$. Call a box *good* if its exponent rectangle lies inside a part whose treatment is not a discard, at distance at least ε from the boundary of that part. Then all its pairs belong to $\sum^*$, and the lemma of that part applies with margin. All other boxes are discarded, and so is the third sum of every good box on which (29) or (31) is used. Each discard leaves a lower bound for $S(\mathcal{A}, X^{1/2})$, since every term of $\sum^*$ is

non-negative and enters (12) with the sign $+$, and every term of a third sum is non-negative and enters (29) or (31) with the sign $+$. For a discarded box, let $\Delta_{P,Q}(\mathcal{B})$ be the sum of $S(\mathcal{B}_{pq}, q)$ over the pairs of $\sum^*$ in the box. For a good box on which (29) or (31) is used, let $\Delta_{P,Q}(\mathcal{B})$ be the third sum of that identity for $\mathcal{E} = \mathcal{B}$, and for the other good boxes let $\Delta_{P,Q}(\mathcal{B}) := 0$. Summing the error terms of (7) over the $\mathcal{O}((\log X)^2)$ good boxes, each $\mathcal{O}(XW^{-1}(\log X)^{-A})$, and using (12) for $\mathcal{B}$, we obtain

$$S(\mathcal{A}, X^{1/2}) \geqslant \frac{2}{\pi W}\Big(S(\mathcal{B}, X^{1/2}) - \sum_{P,Q} \Delta_{P,Q}(\mathcal{B})\Big) - \mathcal{O}\big(XW^{-1}(\log X)^{2-A}\big), \tag{49}$$

where $\sum_{P,Q}$ runs over the dyadic boxes $p \sim P$, $q \sim Q$ of $\sum^*$. In the third sums, split r and s into dyadic ranges, which gives the four-dimensional boxes of Section 8. Call a four-dimensional box *good* if its exponent box lies in G_ε or in L_ε. Lemma 14 gives the sum over every good four-dimensional box an asymptotic formula, so this sum can be moved from $\sum_{P,Q} \Delta_{P,Q}(\mathcal{B})$ to the main term of (49). Every other four-dimensional box stays discarded. There are $\mathcal{O}((\log X)^4)$ four-dimensional boxes, and the errors of the good ones add up to $\mathcal{O}(XW^{-1}(\log X)^{4-A})$.

We bound the discarded sums that remain in (49). The pairs of $\sum^*$ with $|p|^2|q|^4 \geqslant X$ contribute $\mathcal{O}(X/(\log X)^2)$ to the discarded sums over $\mathcal{B}$ (Section 7). Every other pair of a discarded box lies in D, and the box has its exponent rectangle within distance 2ε of a part whose treatment is a discard, of a boundary between parts, or of $\partial\mathsf{D}$. The third sums use the quantities $X^\tau/(PQ)$ and $z(PQ)$, which are constant on a box, in place of $X^\tau/|pq|^2$ and $z(|pq|^2)$. This moves the lower limits of γ and δ in (42) by $\mathcal{O}(1/\log X)$. So a discarded four-dimensional box has its exponent box within distance 2ε of the points of the regions (42) outside $\mathsf{G} \cup \mathsf{L}$, or of the boundary of one of the polytopes whose unions define G, I and L in (16). Let $\Phi_\varepsilon(\theta)$ be the sum of the losses (18) of the 2ε-neighborhoods of these sets, intersected with the closures of D and of the regions (42). Lemma 15, with $K = 2$ and $K = 4$, bounds the other discarded sums over $\mathcal{B}$ by $S(\mathcal{B}, X^{1/2})(\Phi_\varepsilon(\theta) + o(1))$. Its hypotheses $y_i \geqslant c_1$ and $\sum_{i=1}^K y_i \leqslant 1 - c_1$ hold with $c_1 = c(\theta)$ by (43) and (44).

As $\varepsilon \to 0$, these neighborhoods decrease to the closures of the regions in (20), together with finitely many boundaries. The sets G, I, L and P are finite unions of polytopes, so these boundaries are null sets. The integrands are bounded, so $\Phi_\varepsilon(\theta) \to \Phi(\theta)$. Finally $XW^{-1}(\log X)^{4-A} = o(W^{-1}S(\mathcal{B}, X^{1/2}))$ for $A > 5$. Inserting these bounds into (49) gives (48). The same argument applies with P in place of L. □

10 Evaluation of the losses and proof of Theorem 2

The losses (18) are computed numerically by two methods, which differ in the quadrature and in the treatment of G and L, so that each checks the other. Both are implemented in the folder `Code` (see the data availability statement) and use the same limits of the regions and the same tables of Buchstab's function ω of (19) and of its integral $\Omega(u) := \int_1^u \omega(t)\,dt$. On each interval $[k, k+1]$, ω and Ω are Chebyshev series of degree 40, built for $k = 1, 2, \dots$ in turn from (19), which determines ω on $[k, k+1]$ from its values on $[1, k]$. They are tabulated on a grid of step 10^{-5} and evaluated between grid points by cubic Hermite interpolation. By (43), every argument of ω in the losses is below $1/c(\theta) < 40$. On $[1, 3]$, where $\omega(u) = 1/u$ for $u \leqslant 2$ and $\omega(u) = (1 + \log(u-1))/u$ for $u \geqslant 2$, the tables agree with these formulas and their integrals to 10^{-14}. On $[2, 40)$, they satisfy (19) to 1.5×10^{-13}.

In the first method, the integral over β in the two-dimensional losses (18) is evaluated exactly through Ω, since

$$\omega\Big(\frac{1-\alpha-\beta}{\beta}\Big)\frac{1}{\beta^2} = -\frac{1}{1-\alpha}\,\frac{d}{d\beta}\,\Omega\Big(\frac{1-\alpha}{\beta} - 1\Big).$$

With $\alpha + \beta + \gamma$ in place of α and δ in place of β, the same identity gives the integral over δ in the four-dimensional losses. For fixed (α, β, γ), the part of the δ-range outside $\mathsf{G} \cup \mathsf{L}$ is a finite union of intervals, and the identity is applied on each of them. The remaining integrals are computed by composite Gauss–Legendre quadrature. For the regions D_k, the α-axis is cut into 256 equal panels. For the regions E_k, each of the α-, β- and γ-axes is cut into 16 equal panels. The panels are cut further at the points where a limit of integration or the set $\mathsf{G} \cup \mathsf{L}$ changes its formula. On the γ-axis, they are also cut where the δ-range becomes non-empty and where a limit of δ meets $(1 - \alpha - \beta - \gamma)/2$, the value at which the argument of ω equals 1, since the integrand in γ has a kink at these points. Each resulting subinterval carries 10 Gauss–Legendre nodes.

In the second method, the two-dimensional losses are computed by nested adaptive Gauss–Kronrod quadrature of the integrand itself [19], and their error estimate is that of the outer adaptive rule. The four-dimensional losses are computed by scrambled Sobol quasi-Monte Carlo integration [17, 22], with ω read from its table instead of through Ω and with the sets G and L tested pointwise. Each of these losses is the mean of 8 estimates, from independent scramblings of 2^{20} points each. The error estimate, which measures the spread of the 8 estimates, is the square root of the sum of the squared deviations of the 8 estimates from their mean, divided by $(8 \cdot 7)^{1/2}$.

Table 2 lists the losses at $\theta = 0.1129$. The first method gives $\Phi(0.1129) = 0.9990033$, and the second gives 0.9990044. At $\theta = 0.119$, the decomposition of Harman and Lewis, with the regions derived from the lemmas (Remark 27) and with the cut by which Harman and Lewis exclude a subset of G from the region of φ_7 (Appendix E), gives the total 0.9425585, below the bound 0.943 of Harman and Lewis [10, p. 134]. Table 6 compares the losses with the bounds of Harman and Lewis region by region. Table 3 lists the values of θ at which the total loss equals 1 as the groupings are added. Figure 2 shows the total loss of each of these decompositions as a function of θ. At each of the 61 values of θ in the figure, the total loss decreases down the rows of Table 3.

Region	First method	Second method	Error estimate
D_1	0.3201605	0.3201605	$< 10^{-12}$
D_2	0.3437028	0.3437028	$< 10^{-12}$
D_4	0.2132922	0.2132922	$< 10^{-12}$
D_6	0.0480512	0.0480512	$< 10^{-12}$
D_8	0.0684437	0.0684437	$< 10^{-12}$
$\mathsf{E}_3 \setminus (\mathsf{G} \cup \mathsf{L})$	0.0000098	0.0000098	8.8×10^{-10}
$\mathsf{E}_\mathsf{R} \setminus (\mathsf{G} \cup \mathsf{L})$	0.0010206	0.0010207	6.8×10^{-8}
$\mathsf{E}_5 \setminus (\mathsf{G} \cup \mathsf{L})$	0.0006199	0.0006199	4.4×10^{-8}
$\mathsf{E}_7 \setminus (\mathsf{G} \cup \mathsf{L})$	0.0004451	0.0004451	8.5×10^{-9}
$\mathsf{E}_9 \setminus (\mathsf{G} \cup \mathsf{L})$	0.0002727	0.0002726	5.6×10^{-8}
$\mathsf{E}_{10} \setminus (\mathsf{G} \cup \mathsf{L})$	0.0028858	0.0028870	1.0×10^{-6}
$\mathsf{E}_{11} \setminus (\mathsf{G} \cup \mathsf{L})$	0.0000988	0.0000988	1.1×10^{-8}
$\Phi(0.1129)$	0.9990033	0.9990044	1.0×10^{-6}

Table 2: The losses (18) at $\theta = 0.1129$ of the regions of Table 1, with $\mathsf{G} \cup \mathsf{L}$ removed from the regions E_k, computed by the methods of Section 10 and rounded to 7 decimal places, and the error estimate of the second method. The row $\Phi(0.1129)$ gives the sums of the unrounded losses and the square root of the sum of the squared error estimates.

Decomposition	θ with total loss 1	$1/2 - 2\theta$
Harman and Lewis	0.1167825	0.2664351
with R	0.1141073	0.2717854
with R and G	0.1139312	0.2721376
with R, G and P	0.1132049	0.2735902
with R, G and L	0.1128554	0.2742892

Table 3: The value of θ at which the total loss equals 1, for the decomposition of Harman and Lewis with the regions derived from the lemmas (Remark 27) and with the cut by which Harman and Lewis exclude a subset of G from the region of φ_7 (Appendix E), computed by the first method of Section 10. Each row labelled "with" adds the groupings it lists to this decomposition. The column $1/2 - 2\theta$ gives the exponent in $\{\sqrt{p}\} < p^{-(1/2-2\theta)}$.

The computation is not carried out in interval arithmetic, so its accuracy is checked by refining the quadrature of the first method and by comparing the methods. For the first method, the losses at $\theta = 0.1129$ change by at most 2.5×10^{-9}, and their total by 2.6×10^{-10}, between the rule used for Table 2 and the finest

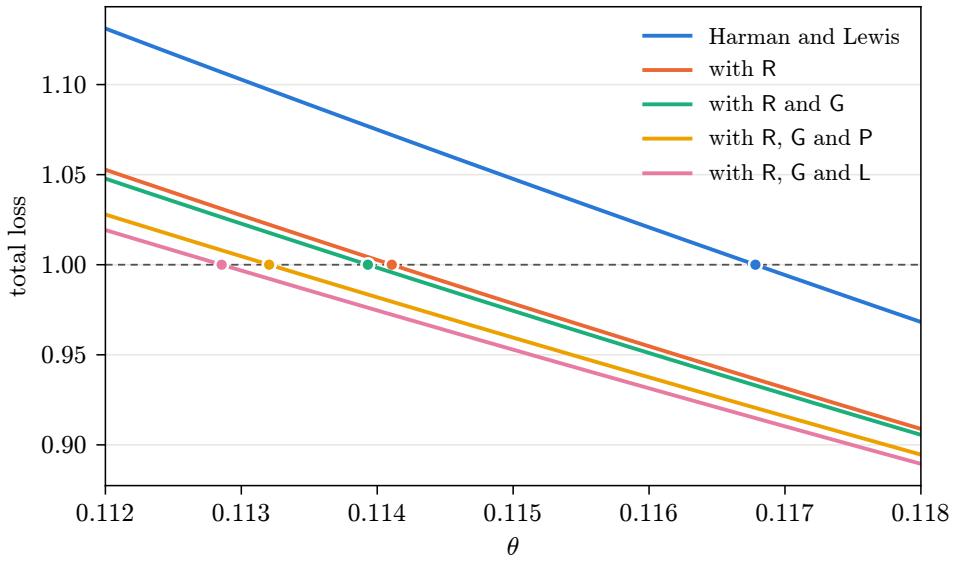


Figure 2: The total loss of each decomposition of Table 3, computed by the first method of Section 10 at 61 values of θ from 0.112 to 0.118. The dashed line marks the total loss 1, and the dots mark the values of θ in Table 3.

rule tested, which has 512 panels on the α-axis for the regions D_k, 32 panels per axis for the regions E_k, and 12 nodes per subinterval. For 8 coarser rules, with 12 to 192 nodes per axis, the total differs from that of the finest rule by amounts that decrease from 8.7×10^{-6} to 6.8×10^{-11} as the number of nodes grows. The totals of the methods differ by 1.2×10^{-6}, which is 1.1 times the error estimate of the second method in Table 2. Both quantities come almost entirely from E_{10}. The gap $1 - \Phi(0.1129) = 9.97 \times 10^{-4}$ exceeds this difference, this error estimate and the change of the total under refinement by a factor of more than 800.

Proof of Theorem 2. Let $\theta = 0.1129$. By Table 2, $\Phi(\theta) < 1$, so Lemma 16 gives $S(\mathcal{A}, X^{1/2}) \gg XW^{-1}/\log X = X^{1/2+\theta}/\log X$, uniformly in ψ. Since $S(\mathcal{A}, X^{1/2})$ is the number of Gaussian primes in $\mathcal{A}$, this is (2). □

With the polytope P of (17) in place of L (Section 3.3), the first and second methods give $\Phi_{\mathsf{P}}(0.1133) = 0.9978243$ and 0.9978250, so the same argument gives (2) with $\theta = 0.1133$, and hence $\{\sqrt{p}\} < p^{-0.2734}$ for infinitely many primes.

11 Conclusion

Of the total loss 0.999 at $\theta = 0.1129$, the regions D_1, D_2 and D_4 account for 0.877. On these regions, after Buchstab's identity (6) introduces a third prime r as in (29), no factorization of pqr satisfies the conditions (10) of the linear estimate once $|r|^2$ is close to $|q|^2$. We have not found another grouping that gives the sums over these regions asymptotic formulas. A further improvement may need new arithmetic information for the Hecke characters λ^m of Appendix A. Harman and Lewis [10, p. 134] call the improvement of their arithmetic information "the more interesting challenge". The numerical evaluation of the losses could be made rigorous by interval arithmetic.

Data availability

The code that computes every number in Section 10 and Appendix D, and the data of Figures 1 and 2, are in the folder `Code` of the repository https://github.com/xuda-ye-math/Gaussian-Prime. The code

is written in Julia 1.13, and the file `Manifest.toml` of the folder fixes the versions of its packages. The file `README.md` of the folder describes each computation and its output.

A Proofs of the linear and bilinear estimates

This appendix derives Lemmas 1 and 2 of Section 4 from the arithmetic information. It begins with standard facts, which Sections 5, 7 and 9 and Appendices B and C also use. Lemma 17 bounds sums over Gaussian primes, and Lemma 18 counts Gaussian integers and their divisors. Lemma 19 is the truncated Perron formula used in the proof of Lemma 2.

Lemma 17 *The number $\pi_G(y)$ of Gaussian primes of norm less than y satisfies $\pi_G(y) = \mathrm{Li}(y) + \mathcal{O}(y\exp(-c(\log y)^{1/2}))$ for some $c > 0$, where $\mathrm{Li}(y) := \int_2^y dt/\log t$. Consequently, for fixed $B \geqslant 0$ and $\kappa_0 \geqslant 1$ and with $H := (\log X)^B$, the following hold once X is large.*

(a) If $Y \geqslant \mu$ and $1 \leqslant \kappa \leqslant \kappa_0$, then

$$\sum_{Y \leqslant |\varpi|^2 < Y(1+1/H)^\kappa} |\varpi|^{-2} \leqslant \frac{2\kappa}{H \log Y}.$$

(b) $\sum_{|\varpi|^2 < y} |\varpi|^{-2} \leqslant \log\log y + C$ for $y \geqslant 3$.

(c) For fixed $a_0 > 0$, uniformly for $a_0 \leqslant a < b \leqslant 1$,

$$\sum_{X^a \leqslant |\varpi|^2 < X^b} |\varpi|^{-2} = \log\frac{b}{a} + \mathcal{O}\Big(\frac{1}{\log X}\Big).$$

(d) $\pi_G(y) - \pi_G(y/2) = (1 + \mathcal{O}(1/\log y))\, y/(2\log y)$.

Proof. For $a \in \{1,3\}$, let $\pi(y;4,a)$ be the number of rational primes $\ell < y$ with $\ell \equiv a \pmod 4$. A Gaussian prime is $1+i$, a prime of norm ℓ for a rational prime $\ell \equiv 1 \pmod 4$, or a rational prime $\ell \equiv 3 \pmod 4$, of norm ℓ^2. Hence $\pi_G(y) = 2\pi(y;4,1) + \pi(y^{1/2};4,3) + \mathcal{O}(1)$. The formula for $\pi_G(y)$ follows by partial summation from the prime number theorem in progressions modulo 4 [23, Section 1.2], in the form

$$\sum_{\ell \leqslant x,\ \ell \equiv a\,(4)} \log \ell = \frac{x}{2}\Big(1 + \mathcal{O}\big(\exp(-c'(\log x)^{3/5}(\log\log x)^{-1/5})\big)\Big), \qquad a \in \{1,3\},$$

with a constant $c' > 0$. Statements (b), (c) and (d) follow by partial summation. In (a), the main term contributes $\log(1+\kappa\log(1+1/H)/\log Y) \leqslant \kappa/(H\log Y)$, and the error term contributes $\mathcal{O}(\exp(-c(\log Y)^{1/2}))$. Since $\log Y \geqslant (\log X)^{4/5}$, the error term is smaller than $1/(H\log Y)$ once X is large, and (a) follows from $\kappa \geqslant 1$. □

Lemma 18 *(a) For $M \geqslant 1$, the number of Gaussian integers u with $X/(2M) \leqslant |u|^2 < X/M$ is at most CX/M. In particular $|\mathcal{B}_u| \leqslant CX/|u|^2$.*

(b) For $k \geqslant 1$ and $Y \geqslant 2$,

$$\sum_{|n|^2 \leqslant Y} d(n)^k \ll_k Y(\log Y)^{2^k}, \qquad \sum_{|n|^2 \leqslant Y} d(n)^k |n|^{-2} \ll_k (\log Y)^{2^k}.$$

(c) A Gaussian integer n has at most $d(n)^{h-1}$ ordered factorizations into h factors.

Proof. For (a), the count is 0 if $X/M < 1$, and $\pi X/(8M) + \mathcal{O}((X/M)^{1/2} + 1)$ otherwise. For (b), d is multiplicative with $d(\varpi^e) = e + 1$. So, by Lemma 17(b), the sum of $d(n)^k |n|^{-2}$ is at most

$$\prod_{|\varpi|^2 \leqslant Y} \sum_{e \geqslant 0} (e+1)^k |\varpi|^{-2e} \leqslant \exp\Big(\sum_{|\varpi|^2 \leqslant Y} \big(2^k |\varpi|^{-2} + C_k |\varpi|^{-4}\big) \Big) \ll_k (\log Y)^{2^k},$$

and the sum of $d(n)^k$ is at most Y times the sum of $d(n)^k |n|^{-2}$. For (c), the number $d_h(n)$ of ordered factorizations satisfies $d_h(n) = \sum_{u|n} d_{h-1}(n/u) \leqslant d(n)^{h-1}$, by induction on h. □

Lemma 19 *For $y > 0$, $c > 0$ and $T_0 \geqslant c$, let*

$$I(y) := \frac{1}{2\pi i} \int_{c - iT_0}^{c + iT_0} y^s \, \frac{ds}{s}.$$

Then $|I(y)| \leqslant y^c \log(3T_0/c)$, and $|I(y) - \mathbf{1}[y > 1]| \leqslant y^c / (\pi T_0 |\log y|)$ for $y \neq 1$.

Proof. The bound $|I(y)| \leqslant y^c \log(3T_0/c)$ follows from $|I(y)| \leqslant (y^c/2\pi) \int_{-T_0}^{T_0} (c^2 + t^2)^{-1/2} \, dt$. For $y > 1$, move the segment to $\operatorname{Re} s = -K$ and let $K \to \infty$. The residue at 0 is 1, the integral over $\operatorname{Re} s = -K$ tends to 0, and each horizontal side is at most

$$\frac{1}{2\pi T_0} \int_{-\infty}^{c} y^u \, du = \frac{y^c}{2\pi T_0 \log y}.$$

For $y < 1$, move the segment to $\operatorname{Re} s = K$, where there is no pole, and obtain the same bound with $|\log y|$. □

For an integer m and a Gaussian integer $n \neq 0$ put $\lambda^m(n) := (n/|n|)^{4m}$, as in Harman and Lewis [10, p. 122]. This is a Hecke character. It is unchanged when n is multiplied by a unit, so it is completely multiplicative under the identification of associates fixed in Section 1. Since $\lambda^m(n) = e^{4im \arg n}$, it detects the argument of n modulo $\pi/2$. Lemmas 20 to 23 below are quoted without proof from Harman and Lewis [10], and Lemma 23 combines their Lemmas 6 and 7. Lemmas 20 and 23 give the linear estimate, Lemma 1. Lemmas 20, 21 and 22 give the bilinear estimate, Lemma 2.

Lemma 20 (Harman and Lewis [10], (12)) *For every $T \geqslant 1$ and all non-negative weights a_n,*

$$\sum_{n \in \mathcal{A}} a_n = \frac{2}{\pi W} \sum_{n \in \mathcal{B}} a_n + \mathcal{O}\Big(\frac{1}{T} \sum_{n \in \mathcal{B}} a_n \Big) + \mathcal{O}\Big(\frac{1}{W} \sum_{m=1}^{T} \Big| \sum_{n \in \mathcal{B}} a_n \lambda^m(n) \Big| \Big).$$

Harman and Lewis state Lemma 20 for any weights and note that it follows from Baker [2, Chapter 2]. It is quoted here for non-negative weights, the only ones used.

Lemma 21 (Harman and Lewis [10], Lemma 1) *For $T \geqslant 1$, $N \geqslant 1$ and all coefficients $c(n)$,*

$$\sum_{m=1}^{T} \Big| \sum_{|n|^2 \leqslant N} c(n) \lambda^m(n) \Big|^2 \ll (T + N) \sum_{|n|^2 \leqslant N}{}' |c_1(n)|^2, \qquad c_1(n) := \sum_{\arg u = \arg n} c(u),$$

where $\sum'$ runs over the primitive Gaussian integers $n = x + iy$, $\gcd(x, y) = 1$.

Lemma 22 (Harman and Lewis [10], Lemma 4) *Let $R, T_0 \geqslant 2$, $T \geqslant S \geqslant R + \exp((\log T T_0)^{4/5})$, $1 \leqslant m \leqslant T$ and $|t| \leqslant T_0$. Then*

$$\Big| \sum_{R \leqslant |\varpi|^2 < S} |\varpi|^{2it} \lambda^m(\varpi) \Big| \ll S \exp\Big(- \frac{\log S}{2 (\log T T_0)^{7/10}} \Big).$$

Lemma 21 is Coleman [5, Lemma 6], and Lemma 22 rests on the zero-free region of Coleman [4].

Lemma 23 (Harman and Lewis [10], Lemmas 6 and 7) *Let $A > 0$, $T := W(\log X)^A$, and let a_u, b_v be divisor-bounded.*

(a) If $U \leqslant X^\tau \mu^{-1}$, then

$$\sum_{m=1}^{T} \Big| \sum_{u\sim U} a_u \sum_{ut\in\mathcal{B}} \lambda^m(ut) \Big| \ll X(\log X)^{-A}.$$

(b) If $U \leqslant X^\zeta \mu^{-1}$, $V \leqslant X^\rho \mu^{-1}$ and $UV \leqslant X^\nu \mu^{-1}$, then

$$\sum_{m=1}^{T} \Big| \sum_{u\sim U} a_u \sum_{v\sim V} b_v \sum_{uvt\in\mathcal{B}} \lambda^m(uvt) \Big| \ll X(\log X)^{-A}.$$

Here t runs over all Gaussian integers with the stated product in $\mathcal{B}$.

Remark 24 in Appendix E explains the conditions $ut \in \mathcal{B}$ and $uvt \in \mathcal{B}$ of Lemma 23.

Proof of Lemma 1. With $c'_n := \sum_{v|n} c_v \geqslant 0$,

$$\sum_v c_v |\mathcal{A}_v| = \sum_{n\in\mathcal{A}} c'_n, \qquad \sum_{n\in\mathcal{B}} c'_n = \sum_v c_v |\mathcal{B}_v|.$$

Apply Lemma 20 with $a_n = c'_n$ and $T = W(\log X)^A$. By Lemma 18, its error term $\mathcal{O}(T^{-1}\sum_{n\in\mathcal{B}} c'_n)$ is at most

$$W^{-1}(\log X)^{-A} \cdot CX \sum_v c_v |v|^{-2} \ll XW^{-1}(\log X)^{C-A}.$$

Its error term with the sums over λ^m is

$$\frac{1}{W} \sum_{m\leqslant T} \Big| \sum_v c_v \sum_{vt\in\mathcal{B}} \lambda^m(vt) \Big|.$$

Under (a), after the range of v is split into $\mathcal{O}(\log X)$ dyadic ranges, it is $\ll XW^{-1}(\log X)^{1-A}$ by Lemma 23(a). Under (b) it is $\ll XW^{-1}(\log X)^{-A}$ by Lemma 23(b). Since A is arbitrary, the lemma follows. □

Proof of Lemma 2. Lemma 19 separates the variables u, v and ϖ. Cauchy's inequality and Lemma 21 then bound the sums over u and over v in mean square over m, and Lemma 22 bounds the sum over ϖ uniformly in m.

Fix A, put $T := W(\log X)^A$, and let E_m be the sum of $a_u b_v \lambda^m(uv\varpi)$ over the triples counted in (21) with $\mathcal{E} = \mathcal{B}$. As in the proof of Lemma 1, it suffices to show that $\sum_{m\leqslant T} |E_m| \ll X(\log X)^{-A}$. A triple occurs only if $X/(8UV) \leqslant |\varpi|^2 < X/(UV)$. Put $Z'_1 := \max(Z_1, X/(8UV))$ and $Z'_2 := \min(Z_2, X/(UV))$, and assume that $Z'_1 < Z'_2$, since otherwise $E_m = 0$. For $N = |uv\varpi|^2 \leqslant 4X$, write $\mathbf{1}[X/2 \leqslant N < X] = \mathbf{1}[X/N > 1] - \mathbf{1}[(X/2)/N > 1]$, and apply Lemma 19 to $y = Y/N$ for $Y \in \{X, X/2\}$, with $c = 1/\log X$ and $T_0 = X^3$. If $|N - Y| \geqslant 1/2$, then $|\log y| \geqslant 1/(16X)$ and the error is at most $16eX/(\pi T_0)$. At most 2 integers N have $|N - Y| < 1/2$, and for them we use $|I(y)| \leqslant e\log(3T_0/c)$. Each $N \leqslant 4X$ is the norm of $X^{o(1)}$ Gaussian integers, and by Lemma 18(c) each of these is the product of $X^{o(1)}$ triples, each with weight $a_u b_v \leqslant X^{o(1)}$. Since λ^m is completely multiplicative, this gives

$$E_m = \frac{1}{2\pi i} \int_{c-iT_0}^{c+iT_0} \big(X^s - (X/2)^s\big) F_m(s) G_m(s) \mathfrak{p}_m(s) \frac{ds}{s} + \mathcal{O}(X^{o(1)}), \tag{50}$$

where

$$F_m(s) := \sum_{u\sim U} a_u \lambda^m(u) |u|^{-2s}, \qquad G_m(s) := \sum_{v\sim V} b_v \lambda^m(v) |v|^{-2s}, \qquad \mathfrak{p}_m(s) := \sum_{Z'_1 \leqslant |\varpi|^2 < Z'_2} \lambda^m(\varpi) |\varpi|^{-2s}.$$

Since $|X^s| = e$ and $\int_{-T_0}^{T_0} |c+it|^{-1}\,dt \ll \log X$, summing (50) over $m \leqslant T$ gives

$$\sum_{m\leqslant T} |E_m| \ll \log X \cdot \sup_{|t|\leqslant T_0} \sum_{m\leqslant T} \big|F_m G_m \mathfrak{p}_m\big|(c+it) + X^{\sigma+o(1)}. \tag{51}$$

By Cauchy's inequality, the sum over m in (51) satisfies

$$\sum_{m\leqslant T} \big|F_m G_m \mathfrak{p}_m\big|(c+it) \leqslant \max_{m\leqslant T} |\mathfrak{p}_m(c+it)| \Big(\sum_{m\leqslant T} |F_m(c+it)|^2\Big)^{1/2} \Big(\sum_{m\leqslant T} |G_m(c+it)|^2\Big)^{1/2}.$$

To bound $\sum_m |F_m|^2$, apply Lemma 21 with $2U$ in place of N and with $c(u) = a_u|u|^{-2s}$ for $u \sim U$. A Gaussian integer u with $\arg u = \arg n$, n primitive, is $u = kn$ with k a positive rational integer. So $|c_1(n)| \leqslant \sum_k d(kn)^{C_0}$, where $k \geqslant 1$ runs over the integers with $k^2|n|^2 \in [U, 2U)$. This sum has at most $(2U)^{1/2}/|n|$ terms. By Cauchy's inequality and $d(kn) \leqslant d(k)d(n)$,

$$\sum_n{}' |c_1(n)|^2 \leqslant (2U)^{1/2} \sum_{k^2\leqslant 2U} d(k)^{2C_0} \sum_{|n|^2<2U/k^2} d(n)^{2C_0}|n|^{-1}. \tag{52}$$

By Lemma 18(b) and partial summation, the sum of $d(n)^{2C_0}|n|^{-1}$ in (52) is $\ll (U/k^2)^{1/2}(\log X)^C$. The remaining sum, of $d(k)^{2C_0}/k$ over the rational integers $k \leqslant (2U)^{1/2}$, is at most a product over the rational primes $\ell \leqslant (2U)^{1/2}$:

$$\prod_{\ell\leqslant (2U)^{1/2}} \sum_{e\geqslant 0} (e+1)^{4C_0}\ell^{-e} \ll (\log X)^{2^{4C_0}}.$$

Here $d(\ell^e) \leqslant (e+1)^2$ for every rational prime ℓ, and $\sum_{\ell\leqslant x} 1/\ell \leqslant \log\log x + \mathcal{O}(1)$ follows from the prime number theorem modulo 4. So the right side of (52) is $\ll U(\log X)^C$, and, since $T \leqslant C_1 U(\log X)^A$,

$$\sum_{m\leqslant T} |F_m|^2 \ll (T+U)U(\log X)^C \ll U^2(\log X)^{A+C}.$$

The same bound holds for G_m with V in place of U.

It remains to bound $\mathfrak{p}_m(c+it)$. The weight $|\varpi|^{-2c}$ in its terms is decreasing in $|\varpi|^2$ and lies in $[e^{-2}, 1]$. By partial summation, $|\mathfrak{p}_m(c+it)|$ is at most 2 times the largest value, over $Z \in [Z_1', Z_2']$, of

$$\Big|\sum_{Z_1'\leqslant|\varpi|^2<Z} \lambda^m(\varpi)|\varpi|^{-2it}\Big|.$$

Each of these sums is the difference of the sums over $2 \leqslant |\varpi|^2 < S$ for $S = Z$ and for $S = Z_1'$. Lemma 22 applies to them with $R = 2$, $|t| \leqslant T_0$ and $1 \leqslant m \leqslant T$, once $T \geqslant S \geqslant 2 + \exp((\log TT_0)^{4/5})$. The condition $S \leqslant T$ holds, since $\theta < 1/6$ gives $S \leqslant X/(UV) \leqslant C_1^2 X^{2\theta} \leqslant T$. For the lower condition on S, $\log(TT_0) \leqslant 4\log X$ and $4^{4/5} < 3.1$ give

$$S \geqslant \frac{\mu^4}{2} > 2 + \exp\big((\log TT_0)^{4/5}\big).$$

Since $S\exp(-\log S/(2(\log TT_0)^{7/10}))$ increases with S and $\log Z_2' \geqslant \log(\mu^4/2)$, Lemma 22 gives

$$\max_m |\mathfrak{p}_m| \ll \frac{X}{UV}\exp\Big(-\frac{3}{4}(\log X)^{1/10}\Big).$$

Collecting the bounds in (51), we obtain

$$\sum_{m\leqslant T} |E_m| \ll X(\log X)^{A+C+1}\exp\Big(-\frac{3}{4}(\log X)^{1/10}\Big) + X^{\sigma+o(1)} \ll X(\log X)^{-A}.$$

□

B Proofs of the sieve lemmas

This appendix proves Lemmas 3, 4 and 5 of Section 4 and Lemma 7 of Section 5. In the proofs of Lemmas 4 and 5, Lemma 3 treats the sifting by the primes of norm below μ^4, and Lemma 2 is applied only to primes of norm at least $\mu^4/2$. As in Section 5, short intervals remove the conditions that couple the variables.

Proof of Lemma 3. The proof is Brun's pure sieve: the inclusion–exclusion over the primes of norm below $z_0(u)$ that divide the cofactor is truncated after the divisors with M prime factors, and the divisors with $M+1$ prime factors bound the error of the truncation. Put $M+1 := \lfloor(\varepsilon_0/4)(\log X)^{1/5}\rfloor$. For $z \geqslant 1$, let $\mathfrak{P}(z)$ be the product of the primes of norm less than z, and for a squarefree $\mathfrak{d}$ let $\mathfrak{o}(\mathfrak{d})$ be its number of prime factors. For squarefree $\mathfrak{d}$ put

$$\mathfrak{b}_{\mathfrak{d}}^{\pm}(u) := \mathbf{1}\big[\mathfrak{d} \mid \mathfrak{P}(z_0(u)),\ \mathfrak{o}(\mathfrak{d}) \leqslant M,\ (-1)^{\mathfrak{o}(\mathfrak{d})} = \pm 1\big], \qquad \mathfrak{b}_{\mathfrak{d}}^{0}(u) := \mathbf{1}\big[\mathfrak{d} \mid \mathfrak{P}(z_0(u)),\ \mathfrak{o}(\mathfrak{d}) = M+1\big],$$

and let $\mathfrak{b}_{\mathfrak{d}}^{+}(u)$, $\mathfrak{b}_{\mathfrak{d}}^{-}(u)$ and $\mathfrak{b}_{\mathfrak{d}}^{0}(u)$ be 0 for $\mathfrak{d}$ not squarefree. We first show that $\sum_u c_u S'(\mathcal{E}_u, z_0(u))$ has an asymptotic formula if, for each $\mathfrak{b} \in \{\mathfrak{b}^+, \mathfrak{b}^-, \mathfrak{b}^0\}$, the sum $\sum_{u,\mathfrak{d}} c_u \mathfrak{b}_{\mathfrak{d}}(u)|\mathcal{E}_{u\mathfrak{d}}|$ has one.

For $u \mid n$, let $\mathfrak{n}$ be the number of distinct primes of norm less than $z_0(u)$ that divide n/u. The number of $\mathfrak{d} \mid \mathfrak{P}(z_0(u))$ with $\mathfrak{d} \mid n/u$ and $\mathfrak{o}(\mathfrak{d}) = i$ is $\binom{\mathfrak{n}}{i}$. For $\mathfrak{n} \geqslant 1$,

$$\sum_{i=0}^{M} (-1)^i \binom{\mathfrak{n}}{i} = (-1)^M \binom{\mathfrak{n}-1}{M},$$

and this is at most $\binom{\mathfrak{n}}{M+1}$ in absolute value. Hence

$$S'(\mathcal{E}_u, z_0(u)) = \sum_{\mathfrak{d}} \big(\mathfrak{b}_{\mathfrak{d}}^{+}(u) - \mathfrak{b}_{\mathfrak{d}}^{-}(u)\big)|\mathcal{E}_{u\mathfrak{d}}| + \Delta_u, \qquad |\Delta_u| \leqslant \sum_{\mathfrak{d}} \mathfrak{b}_{\mathfrak{d}}^{0}(u)|\mathcal{E}_{u\mathfrak{d}}|. \tag{53}$$

Multiply by c_u and sum over u. By assumption, the sums of $c_u(\mathfrak{b}_{\mathfrak{d}}^{+}(u) - \mathfrak{b}_{\mathfrak{d}}^{-}(u))|\mathcal{A}_{u\mathfrak{d}}|$ in (53) over u and $\mathfrak{d}$ are $2/(\pi W)$ times those for $\mathcal{B}$, up to $\mathcal{O}(XW^{-1}(\log X)^{-A})$. The remainders Δ_u contribute at most

$$\sum_{u,\mathfrak{d}} c_u \mathfrak{b}_{\mathfrak{d}}^{0}(u)\Big(|\mathcal{A}_{u\mathfrak{d}}| + \frac{2}{\pi W}|\mathcal{B}_{u\mathfrak{d}}|\Big) = \frac{4}{\pi W}\sum_{u,\mathfrak{d}} c_u \mathfrak{b}_{\mathfrak{d}}^{0}(u)|\mathcal{B}_{u\mathfrak{d}}| + \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big),$$

where the equality holds by assumption. By Lemma 18 and Lemma 17(b), with $\sum_{|\varpi|^2<\mu^4} |\varpi|^{-2} \leqslant \log\log X$,

$$\sum_{u,\mathfrak{d}} c_u \mathfrak{b}_{\mathfrak{d}}^{0}(u)|\mathcal{B}_{u\mathfrak{d}}| \leqslant CX\sum_{u} c_u|u|^{-2}\frac{(\log\log X)^{M+1}}{(M+1)!} \ll X(\log X)^C\Big(\frac{e\log\log X}{M+1}\Big)^{M+1} \leqslant X(\log X)^C e^{-(M+1)},$$

once $M+1 \geqslant e^2\log\log X$. Finally $e^{-(M+1)} \leqslant \exp(-(\varepsilon_0/5)(\log X)^{1/5})$ is smaller than every power of $1/\log X$.

It remains to show that the assumption holds in cases (a) and (b). The indicators $\mathfrak{b}_{\mathfrak{d}}(u)$ vanish unless $|\mathfrak{d}|^2 < \mu^{4(M+1)} \leqslant X^{\varepsilon_0}$. In case (a), $\sum_{v=u\mathfrak{d}} c_u \mathfrak{b}_{\mathfrak{d}}(u)$ is non-negative, at most $d(v)^{C_0+1}$, and supported on $|v|^2 \leqslant X^{\tau}\mu^{-1}$, so Lemma 1(a) applies. In case (b), $\mathfrak{b}_{\mathfrak{d}}$ does not depend on u, since z_0 is constant. After the range of $u_2\mathfrak{d}$ is split into dyadic ranges, Lemma 1(b) applies with $u_2\mathfrak{d}$ as its variable u'. □

The proofs of Lemmas 4 and 5 confine primes to the short intervals J_λ of Section 5, with the indices λ_q defined there. The following estimates bound the terms in which two primes share a short interval. For an index λ put $\mathfrak{s}_\lambda := \sum_{|\varpi|^2\in J_\lambda} |\varpi|^{-2}$. If $\xi_{\lambda+1} > \mu^4$, then $\xi_\lambda \geqslant \mu^4/2 \geqslant \mu$, so Lemma 17(a) gives

$$\mathfrak{s}_\lambda \leqslant \frac{2}{H\log(\mu^4/2)}. \tag{54}$$

In sums over λ, the index runs over the λ with $J_\lambda \cap [1, 2X] \neq \emptyset$. By Lemma 17(b), $\sum_\lambda \mathfrak{s}_\lambda \leqslant 2\log\log X$, so that

$$\sum_{\lambda:\ \xi_{\lambda+1}>\mu^4} \mathfrak{s}_\lambda^2 \leqslant \frac{4\log\log X}{H\log(\mu^4/2)}. \tag{55}$$

Proof of Lemma 4. Buchstab's identity (6) lowers the level $l(n)$ to μ^4, where Lemma 3 applies. Once the prime q of the subtracted sum is confined to a short interval, that sum takes the form (21), up to terms that (54) and (55) make small. Fix A, let C be such that $\sum_{|n|^2\leqslant X} a_n|n|^{-2} \leqslant (\log X)^C$ (Lemma 18), and take $B = A + C + 3$. Split the support of a into $\mathcal{N}_1 := \{n : l(n) \leqslant \mu^4\}$ and $\mathcal{N}_2 := \{n : l(n) > \mu^4\}$. On $\mathcal{N}_1$, Lemma 3(a) applies with $c_n = a_n \mathbf{1}_{\mathcal{N}_1}(n)$, the level $z_0 = l$ and $\varepsilon_0 = \varepsilon/2$, since $X^{\tau-\varepsilon} \leqslant X^{\tau-\varepsilon/2}\mu^{-1}$. On $\mathcal{N}_2$, Buchstab's identity (6) gives

$$S'(\mathcal{E}_n, l(n)) = S'(\mathcal{E}_n, \mu^4) - \sum_{\mu^4\leqslant|q|^2<l(n)} S'(\mathcal{E}_{nq}, q), \tag{56}$$

and the term $S'(\mathcal{E}_n, \mu^4)$ of (56), summed with a_n, has an asymptotic formula by Lemma 3(a) at the constant level μ^4. It remains to treat

$$\Sigma_0(\mathcal{E}) := \sum_{n\in\mathcal{N}_2} a_n \sum_{\mu^4\leqslant|q|^2<l(n)} S'(\mathcal{E}_{nq}, q).$$

Two conditions couple n, q and the cofactor: $|q|^2 < l(n)$, and the condition that every prime factor of the cofactor is $\succcurlyeq q$. For $n \in \mathcal{N}_2$ let $\lambda(n)$ be the index with $l(n) \in J_{\lambda(n)}$. Then

$$\mathbf{1}[|q|^2 < l(n)] = \mathbf{1}[\lambda_q < \lambda(n)] + \mathbf{1}[\lambda_q = \lambda(n)]\,\mathbf{1}[|q|^2 < l(n)].$$

Moreover, a prime factor $\varpi' \prec q$ of the cofactor with norm at least ξ_{λ_q} has its norm in J_{λ_q}, so

$$0 \leqslant S'(\mathcal{E}_{nq}, \xi_{\lambda_q}) - S'(\mathcal{E}_{nq}, q) \leqslant \sum_{|\varpi'|^2\in J_{\lambda_q}} |\mathcal{E}_{nq\varpi'}|.$$

With q running over the primes of norm at least μ^4 in J_λ, this gives

$$\Big|\Sigma_0(\mathcal{E}) - \sum_\lambda \mathfrak{M}_\lambda(\mathcal{E})\Big| \leqslant \sum_\lambda \big(\mathfrak{D}_\lambda(\mathcal{E}) + \mathfrak{D}'_\lambda(\mathcal{E})\big), \tag{57}$$

where

$$\mathfrak{M}_\lambda(\mathcal{E}) := \sum_{\lambda(n)>\lambda} a_n \sum_q S'(\mathcal{E}_{nq}, \xi_\lambda), \quad \mathfrak{D}_\lambda(\mathcal{E}) := \sum_{\lambda(n)=\lambda} a_n \sum_q |\mathcal{E}_{nq}|, \quad \mathfrak{D}'_\lambda(\mathcal{E}) := \sum_{\lambda(n)\geqslant\lambda} a_n \sum_q \sum_{|\varpi'|^2\in J_\lambda} |\mathcal{E}_{nq\varpi'}|.$$

Each of these sums has the form

$$\sum_{u,v} b'_u b_v \,\#\{q : |q|^2 \in J_\lambda,\ |q|^2 \geqslant \mu^4,\ uvq \in \mathcal{E}\},$$

with $u = n$, with b'_u equal to a_n restricted to $\lambda(n) > \lambda$, $\lambda(n) = \lambda$ or $\lambda(n) \geqslant \lambda$, and with v the cofactor of nq. For $\mathfrak{M}_\lambda$, the coefficient b_v is the indicator that every prime factor of v has norm at least ξ_λ. For $\mathfrak{D}_\lambda$ it is 1, and for $\mathfrak{D}'_\lambda$ it is the number of primes in J_λ dividing v, which is at most $d(v)$. For $\lambda(n) \geqslant \lambda$ and $|q|^2 \in J_\lambda$,

$$|q|^2 < \xi_{\lambda+1} \leqslant (1 + 1/H)\, l(n) \leqslant \frac{2C_l X^\tau}{|n|^2},$$

so $|v|^2 \geqslant (X/2)/|nq|^2 \geqslant W/(4C_l)$. In addition, $|n|^2 \geqslant W/C_l$ and $|q|^2 \geqslant \mu^4$. After u and v are split into dyadic ranges, Lemma 2 gives $\mathfrak{M}_\lambda$, $\mathfrak{D}_\lambda$ and $\mathfrak{D}'_\lambda$ asymptotic formulas, uniformly in λ. At most $2H\log X$ indices and $\mathcal{O}((\log X)^2)$ pairs of dyadic ranges occur, so Lemma 2 is applied $\mathcal{O}(H(\log X)^3)$ times. With the saving $(\log X)^{-A-B-5}$ in each application, the errors add up to $\mathcal{O}(XW^{-1}(\log X)^{-A})$.

By (57), $\Sigma_0(\mathcal{A}) - \frac{2}{\pi W}\Sigma_0(\mathcal{B})$ is $\mathcal{O}(XW^{-1}(\log X)^{-A})$ plus

$$\mathcal{O}\Big(\sum_\lambda (\mathfrak{D}_\lambda + \mathfrak{D}'_\lambda)(\mathcal{A}) + \frac{2}{\pi W}\sum_\lambda (\mathfrak{D}_\lambda + \mathfrak{D}'_\lambda)(\mathcal{B})\Big),$$

and the sums over $\mathcal{A}$ equal $\frac{2}{\pi W}$ times those over $\mathcal{B}$ up to the same error. By Lemma 18,

$$\sum_\lambda \mathfrak{D}_\lambda(\mathcal{B}) \leqslant CX \sum_n a_n|n|^{-2}\mathfrak{s}_{\lambda(n)}, \qquad \sum_\lambda \mathfrak{D}'_\lambda(\mathcal{B}) \leqslant CX \sum_n a_n|n|^{-2} \sum_\lambda \mathfrak{s}_\lambda^2,$$

and, by (54) and (55), both are $\ll X(\log X)^C \log\log X/(H(\log X)^{4/5}) \ll X(\log X)^{-A}$. Hence Σ_0 has an asymptotic formula, and so does $\sum_n a_n S'(\mathcal{E}_n, l(n))$. □

Proof of Lemma 5. Here $|n|^2$ may be much smaller than W, while Lemma 4 needs $|n|^2 \geqslant W/C_l$. So Buchstab's identity (6) is iterated until the product of n and the primes it introduces has norm at least W, and Lemma 4 is applied to that product. Fix A, let C be as in the proof of Lemma 4, and take $B = A+C+4$. The set $\mathcal{N}_1 := \{n : l(n) \leqslant \mu^4\}$ is handled by Lemma 3(a) with the level l, since $|n|^2 < W \leqslant X^{\tau-2\theta}$. On $\mathcal{N}_2 := \{n : l(n) > \mu^4\}$, (56) holds, and the term $S'(\mathcal{E}_n, \mu^4)$ of (56) again has an asymptotic formula by Lemma 3(a). It remains to treat the sum $\Sigma_0(\mathcal{E})$, defined as in the proof of Lemma 4.

A *chain of length* $i \geqslant 0$ is a tuple $\mathfrak{c} = (n, p_1, \dots, p_i, q)$ with $n \in \mathcal{N}_2$ and primes $p_1 \succ \cdots \succ p_i \succ q$ of norm at least μ^4, such that the largest of these primes (p_1, or q if $i = 0$) has norm less than $l(n)$, and $|np_1 \cdots p_{i'}|^2 < W$ for $1 \leqslant i' \leqslant i$. Write $N(\mathfrak{c}) := np_1 \cdots p_i q$ and $q(\mathfrak{c}) := q$. Let $\mathcal{K}_i$ be the set of chains of length i, split into the exits $\mathcal{K}_i^{\geqslant}$, with $|N(\mathfrak{c})|^2 \geqslant W$, and the continuations $\mathcal{K}_i^{<}$, with $|N(\mathfrak{c})|^2 < W$, and put

$$\Sigma_i(\mathcal{E}) := \sum_{\mathfrak{c}\in\mathcal{K}_i} a_n S'(\mathcal{E}_{N(\mathfrak{c})}, q(\mathfrak{c})).$$

For $\mathfrak{c} \in \mathcal{K}_i^{<}$, Buchstab's identity (6) gives

$$S'(\mathcal{E}_{N(\mathfrak{c})}, q) = S'(\mathcal{E}_{N(\mathfrak{c})}, \mu^4) - \sum_{q'\prec q,\ |q'|^2\geqslant\mu^4} S'(\mathcal{E}_{N(\mathfrak{c})q'}, q').$$

The map $(\mathfrak{c}, q') \mapsto (n, p_1, \dots, p_i, q, q')$, on the pairs that occur in these sums, is a bijection onto $\mathcal{K}_{i+1}$. Hence $\Sigma_i = \mathfrak{E}_i + \mathfrak{F}_i - \Sigma_{i+1}$, where

$$\mathfrak{E}_i(\mathcal{E}) := \sum_{\mathfrak{c}\in\mathcal{K}_i^{\geqslant}} a_n S'(\mathcal{E}_{N(\mathfrak{c})}, q(\mathfrak{c})), \qquad \mathfrak{F}_i(\mathcal{E}) := \sum_{\mathfrak{c}\in\mathcal{K}_i^{<}} a_n S'(\mathcal{E}_{N(\mathfrak{c})}, \mu^4).$$

A chain of length i has $\mu^{4i} \leqslant |p_1 \cdots p_i|^2 < W$, so $\mathcal{K}_i$ is empty for $i > i_{\max} := \lfloor \sigma(\log X)^{1/5}/4 \rfloor$. Hence, for $\mathcal{E} = \mathcal{A}$ and for $\mathcal{E} = \mathcal{B}$,

$$\Sigma_0 = \sum_{i=0}^{i_{\max}} (-1)^i (\mathfrak{E}_i + \mathfrak{F}_i). \tag{58}$$

The primes of a chain are distinct and their order is forced, so a chain is determined by its product and the set of its primes. The number of chains with a given product N is therefore at most $d(N)$, and every coefficient defined from chains in this proof is at most $d(N)^{C_0+1}$, uniformly in i. The sum $\mathfrak{F}_i$ is $\sum_N c_i(N) S'(\mathcal{E}_N, \mu^4)$, where $c_i(N)$ is the sum of a_n over the chains $\mathfrak{c} \in \mathcal{K}_i^{<}$ with $N(\mathfrak{c}) = N$. This $c_i(N)$ is supported on $|N|^2 < W$, so Lemma 3(a) gives $\mathfrak{F}_i$ an asymptotic formula, uniformly in i.

For the exits, Lemma 4 is applied with $N(\mathfrak{c})$ in place of n, and (59) below gives its hypotheses. Let $j_{\max} := \lceil \sigma/\theta \rceil$ and $C_E := 2C_l^{j_{\max}+2}$. We claim that every $\mathfrak{c} \in \mathcal{K}_i^{\geqslant}$ satisfies

$$W \leqslant |N(\mathfrak{c})|^2 \leqslant C_E X^{\tau-\theta/2}, \qquad |N(\mathfrak{c})|^2 |q(\mathfrak{c})|^2 \leqslant C_E X^{\tau}. \tag{59}$$

Write the primes of $\mathfrak{c}$ as $p_1 \succ \cdots \succ p_k$ with $p_k = q$ and $k = i+1$, and put $N' := N(\mathfrak{c})/q$, so that $|N'|^2 < W \leqslant |N(\mathfrak{c})|^2$. Let $j = j(|n|^2)$. Then $0 \leqslant j < \sigma/\theta \leqslant j_{\max}$. Since $|n|^2 \geqslant WX^{-(j+1)\theta}$, we have $z(|n|^2) \leqslant X^{(j+3)\theta/(j+2)} \leqslant X^{3\theta/2}$, and

$$|n|^2 |p_1|^{2(j+2)} < |n|^2 l(n)^{j+2} \leqslant C_l^{j+2} X^{\tau}.$$

We bound $|N(\mathfrak{c})|^2|q|^2$ first. If $|q|^2 \leqslant X^{\theta}$, then $|N(\mathfrak{c})|^2|q|^2 = |N'|^2|q|^4 < WX^{2\theta} = X^{\tau}$. If $|q|^2 > X^{\theta}$, then every prime p_h of $\mathfrak{c}$ has norm greater than X^{θ}, so

$$|n|^2 X^{(k-1)\theta} \leqslant |N'|^2 < W \leqslant |n|^2 X^{(j+1)\theta}.$$

This gives $k \leqslant j+1$, and then

$$|N(\mathfrak{c})|^2|q|^2 \leqslant |n|^2|p_1|^{2(k+1)} \leqslant |n|^2|p_1|^{2(j+2)} \leqslant C_l^{j+2} X^{\tau}.$$

It remains to bound $|N(\mathfrak{c})|^2$. Since $|N'|^2 < W$ and $|q|^2 \leqslant |p_1|^2 < l(n) \leqslant C_l X^{3\theta/2}$,

$$|N(\mathfrak{c})|^2 < W|q|^2 < C_l W X^{3\theta/2} = C_l X^{\tau-\theta/2}.$$

Now confine the prime $q(\mathfrak{c})$ to short intervals. For $\mathfrak{c} \in \mathcal{K}_i^{\geqslant}$ with $\lambda_{q(\mathfrak{c})} = \lambda$, the difference $S'(\mathcal{E}_{N(\mathfrak{c})}, \xi_\lambda) - S'(\mathcal{E}_{N(\mathfrak{c})}, q(\mathfrak{c}))$ lies between 0 and $\sum_{|\varpi'|^2 \in J_\lambda} |\mathcal{E}_{N(\mathfrak{c})\varpi'}|$. Hence

$$\Big|\mathfrak{E}_i - \sum_\lambda \sum_N c_{i,\lambda}(N) S'(\mathcal{E}_N, \xi_\lambda)\Big| \leqslant \mathfrak{E}_i' = \sum_\lambda \sum_N c_{i,\lambda}(N) \sum_{|\varpi'|^2 \in J_\lambda} |\mathcal{E}_{N\varpi'}|, \tag{60}$$

where $c_{i,\lambda}(N)$ is the sum of a_n over the chains $\mathfrak{c} \in \mathcal{K}_i^{\geqslant}$ with $N(\mathfrak{c}) = N$ and $\lambda_{q(\mathfrak{c})} = \lambda$. By (59), on the support of $c_{i,\lambda}$ we have $W \leqslant |N|^2 \leqslant C_E X^{\tau-\theta/2}$ and $\xi_\lambda \leqslant |q|^2 \leqslant C_E X^\tau/|N|^2$. So Lemma 4, with the constant level ξ_λ, with $\theta/4$ in place of ε and with C_E in place of C_l, gives each sum $\sum_N c_{i,\lambda}(N) S'(\mathcal{E}_N, \xi_\lambda)$ in (60) an asymptotic formula. The sum $\mathfrak{E}_i'$ is of the form (21) with $u = N$, with the prime ϖ' and with v the cofactor. The norm of ϖ' lies in J_λ and is at least $\mu^4/2$, and $|v|^2 \geqslant (X/2)/(|N|^2\xi_{\lambda+1}) \geqslant W/(3C_E)$. So Lemma 2 gives it an asymptotic formula.

The *stage* i consists of the sums $\mathfrak{E}_i$ and $\mathfrak{F}_i$. Each stage has at most $2H \log X$ indices λ, at most $\log X$ dyadic ranges of N and at most $\log X$ dyadic ranges of v. So (58) and (60) express Σ_0 through at most $2H(\log X)^2$ applications of Lemma 4, at most $2H(\log X)^3$ of Lemma 2 and 1 of Lemma 3 per stage, over at most $(\log X)^{1/5}$ stages. These are $\mathcal{O}(H(\log X)^{16/5})$ sums in all. With the saving $(\log X)^{-A-B-5}$ in each, their errors add up to $\mathcal{O}(XW^{-1}(\log X)^{-A})$.

It remains to bound the remainders $\mathfrak{E}_i'$. By Lemma 18,

$$\sum_i \mathfrak{E}_i'(\mathcal{B}) \leqslant CX \sum_n a_n |n|^{-2} \sum_{i \geqslant 0} \sum_{p_1 \succ \cdots \succ p_i} \prod_h |p_h|^{-2} \sum_\lambda \mathfrak{s}_\lambda^2,$$

where the factor $\mathfrak{s}_\lambda^2$ comes from the primes q and ϖ', which both lie in J_λ. The sum of $\prod_h |p_h|^{-2}$ over the tuples $p_1 \succ \cdots \succ p_i$ of primes of norm at most X is at most $(2 \log\log X)^i/i!$, so the sum over i is at most $(\log X)^2$. With (55),

$$\sum_i \mathfrak{E}_i'(\mathcal{B}) \ll \frac{X(\log X)^{C+2} \log\log X}{H(\log X)^{4/5}} \ll X(\log X)^{-A},$$

and the asymptotic formulas for the sums $\mathfrak{E}_i'$ give $\sum_i \mathfrak{E}_i'(\mathcal{A}) \ll XW^{-1}(\log X)^{-A}$. Summing (58) term by term gives

$$\Sigma_0(\mathcal{A}) - \frac{2}{\pi W} \Sigma_0(\mathcal{B}) = \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big).$$

□

Proof of Lemma 7. The proof compares $\Sigma_{\mathcal{C}}$, tuple by tuple, with the sum of the pieces $\mathcal{P}_{\boldsymbol{\lambda}}(\mathcal{E}; y_{\boldsymbol{\lambda}})$ of (22) over the index vectors that satisfy all constraints. The difference is bounded by the straddling pieces and the window pieces, each of which confines a prime to a short interval. Fix A and take $B = A+1$. For a tuple $\boldsymbol{\varpi}$, let $\boldsymbol{\lambda}(\boldsymbol{\varpi}) := (\lambda_{\varpi_1}, \dots, \lambda_{\varpi_k})$ be its index vector. For a constraint c, let $g_c(\boldsymbol{\lambda}) := 1$ if $\boldsymbol{\lambda}$ satisfies c and $h_c(\boldsymbol{\lambda}) := 1$ if $\boldsymbol{\lambda}$ straddles c, and let both be 0 otherwise. An index vector that neither satisfies nor straddles c has no tuple satisfying c, so

$$\mathbf{1}_c(\boldsymbol{\varpi}) = g_c(\boldsymbol{\lambda}(\boldsymbol{\varpi})) + \mathbf{1}_c(\boldsymbol{\varpi})\, h_c(\boldsymbol{\lambda}(\boldsymbol{\varpi})), \qquad g_c h_c = 0.$$

Let $\Xi_{\boldsymbol{\lambda}}(t)$ be the indicator that every prime factor of t has norm at least the level $y_{\boldsymbol{\lambda}}$ of (23). For $\boldsymbol{\lambda} = \boldsymbol{\lambda}(\boldsymbol{\varpi})$ we have $\Xi(\boldsymbol{\varpi}, t) \leqslant \Xi_{\boldsymbol{\lambda}}(t)$, and the difference is at most the number of prime factors of t with norm in $J_{\lambda'}$ for some λ' in the window $\mathcal{W}(\boldsymbol{\lambda})$ of (24). Indeed, in type (S), a prime factor $\varpi' \prec \varpi_{i_s}$ of t with $|\varpi'|^2 \geqslant \xi_{\lambda_{i_s}}$ has its norm in $J_{\lambda_{i_s}}$. In type (M), a prime factor ϖ' of t with $\inf_{\mathcal{T}(\boldsymbol{\lambda})} m_s \leqslant |\varpi'|^2 < m_s(\boldsymbol{\varpi})$ has its norm in $J_{\lambda'}$ for some $\lambda' \in \mathcal{W}(\boldsymbol{\lambda})$. In types (C) and (F), $\Xi_{\boldsymbol{\lambda}} = \Xi$. All these indicators take values in $\{0, 1\}$, so for $\boldsymbol{\lambda} = \boldsymbol{\lambda}(\boldsymbol{\varpi})$,

$$\Big|\prod_c \mathbf{1}_c(\boldsymbol{\varpi})\, \Xi(\boldsymbol{\varpi}, t) - \prod_c g_c(\boldsymbol{\lambda})\, \Xi_{\boldsymbol{\lambda}}(t)\Big| \leqslant \sum_c h_c(\boldsymbol{\lambda}) + \prod_c g_c(\boldsymbol{\lambda}) \big(\Xi_{\boldsymbol{\lambda}}(t) - \Xi(\boldsymbol{\varpi}, t)\big). \tag{61}$$

Let $\mathfrak{I}_0$ be the set of index vectors that satisfy all constraints, and $\mathfrak{I}_c$ the set of those that straddle c. Multiply (61) by $\mathbf{1}[\varpi_1 \cdots \varpi_k t \in \mathcal{E}] \prod_i \mathbf{1}[|\varpi_i|^2 \in \mathcal{I}_i]$ and sum. This gives

$$\Big|\Sigma_{\mathcal{C}}(\mathcal{E}) - \sum_{\boldsymbol{\lambda} \in \mathfrak{I}_0} \mathcal{P}_{\boldsymbol{\lambda}}(\mathcal{E}; y_{\boldsymbol{\lambda}})\Big| \leqslant E^+(\mathcal{E}), \tag{62}$$

where

$$E^+(\mathcal{E}) := \sum_c \sum_{\boldsymbol{\lambda} \in \mathfrak{I}_c} \mathcal{P}_{\boldsymbol{\lambda}}(\mathcal{E}; 1) + \sum_{\boldsymbol{\lambda} \in \mathfrak{I}_0} \sum_{\lambda' \in \mathcal{W}(\boldsymbol{\lambda})} \mathcal{P}'_{\boldsymbol{\lambda}, \lambda'}(\mathcal{E}).$$

At most $(2H \log X)^k$ index vectors have $\mathcal{T}(\boldsymbol{\lambda}) \neq \emptyset$. By (62) for $\mathcal{A}$ and for $\mathcal{B}$, and by the hypotheses with $A' = A + (k+1)(A+3)$ in place of A, we obtain

$$\Sigma_{\mathcal{C}}(\mathcal{A}) - \frac{2}{\pi W}\Sigma_{\mathcal{C}}(\mathcal{B}) = \mathcal{O}\Big(E^+(\mathcal{A}) + \frac{2}{\pi W}E^+(\mathcal{B})\Big) + \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big),$$
$$E^+(\mathcal{A}) = \frac{2}{\pi W}E^+(\mathcal{B}) + \mathcal{O}\big(XW^{-1}(\log X)^{-A}\big).$$

It therefore suffices to show

$$E^+(\mathcal{B}) \ll \frac{X(\log\log X)^k}{H(\log X)^{4/5}}. \tag{63}$$

By Lemma 18(a), a tuple $\boldsymbol{\varpi}$ contributes at most $CX/|\varpi_1 \cdots \varpi_k|^2$ to $\mathcal{P}_{\boldsymbol{\lambda}}(\mathcal{B}; 1)$, and a tuple $(\boldsymbol{\varpi}, \varpi')$ at most $CX/|\varpi_1 \cdots \varpi_k \varpi'|^2$ to $\mathcal{P}'_{\boldsymbol{\lambda}, \lambda'}(\mathcal{B})$. If $\boldsymbol{\lambda}$ straddles c, every tuple of $\mathcal{T}(\boldsymbol{\lambda})$ has $m_c(\boldsymbol{\varpi})$ within the factor $(1+1/H)^{2k}$ of Z_c. Choose i_1 with $e_{c,i_1} \neq 0$. For fixed ϖ_i, $i \neq i_1$, the norm $|\varpi_{i_1}|^2$ then lies in an interval $[Y, Y(1+1/H)^{4k})$ with $Y \geqslant \mu^4$, and Lemma 17(a) bounds the sum of $|\varpi_{i_1}|^{-2}$ over it by $8k/(H \log \mu^4)$. Each other variable contributes at most $\sum_{|\varpi|^2 \leqslant X} |\varpi|^{-2} \leqslant 2 \log\log X$. This bounds the sums over $\mathfrak{I}_c$ in $E^+(\mathcal{B})$ by the right side of (63). In the sum over $\mathfrak{I}_0$, for fixed $\boldsymbol{\lambda}$, the norm of ϖ' runs over an interval $[Y', Y'(1+1/H)^{2k+2})$ with $Y' \geqslant \mu^4/2$, and Lemma 17(a) bounds the sum of $|\varpi'|^{-2}$ over it by $(4k+4)/(H \log(\mu^4/2))$. The remaining sums over $\mathcal{T}(\boldsymbol{\lambda})$ add up, over all $\boldsymbol{\lambda}$, to at most $(2 \log\log X)^k$. This bounds the sum over $\mathfrak{I}_0$ by the right side of (63).

The terms of $\Sigma_{\mathcal{C}}$ in which $\prec$ compares two primes of equal norm need no separate treatment: (63) covers those in which two of the ϖ_i have equal norm, and those in which a prime factor of t has the norm of ϖ_{i_s}. Two primes of equal norm lie in the same short interval, so the index vector of such a tuple straddles the order constraint between them. A prime factor of t with the norm of ϖ_{i_s} has its norm in $J_{\lambda'}$ with $\lambda' = \lambda_{i_s} \in \mathcal{W}(\boldsymbol{\lambda})$. □

C Discarded sums as integrals

This appendix proves Lemma 15, which the proof of Lemma 16 uses to bound the discarded sums over $\mathcal{B}$. The proof counts tuples of primes in (64), writes the left side of (47) as an integral of a function g that sums over the factorizations of the cofactor, and identifies g with Buchstab's function (19) through the recursion (65).

Proof of Lemma 15. We first count tuples $(\varpi_0, \dots, \varpi_{K'})$ with a *free prime* ϖ_0, restricted only by $\varpi_0 \cdots \varpi_{K'} \in \mathcal{B}$. Let $K' \geqslant 1$, and let $\mathcal{R}' \subset \mathbb{R}^{K'}$ be bounded, with boundary of measure 0, and with $y_i \geqslant c_1$ and $\sum_i y_i \leqslant 1 - c_1$ on $\mathcal{R}'$. We claim

$$\#\big\{(\varpi_0, \dots, \varpi_{K'}) : \boldsymbol{y}(\varpi_1, \dots, \varpi_{K'}) \in \mathcal{R}',\ \varpi_0 \cdots \varpi_{K'} \in \mathcal{B}\big\} = \frac{X}{2 \log X}\Big(\int_{\mathcal{R}'} \frac{d\boldsymbol{y}}{y_1 \cdots y_{K'}(1 - \sum_i y_i)} + o(1)\Big). \tag{64}$$

For fixed $\varpi_1, \dots, \varpi_{K'}$ with $Y = \prod_i |\varpi_i|^2 \leqslant X^{1-c_1}$, Lemma 17(d) gives the number of ϖ_0 as

$$\pi_G(X/Y) - \pi_G(X/(2Y)) = (1 + o(1))\frac{X}{2Y(1 - \sum_i y_i) \log X}.$$

So the left side of (64) is

$$(1+o(1))\frac{X}{2\log X}\sum_{\varpi_1,\dots,\varpi_{K'}} F(\boldsymbol{y}(\boldsymbol{\varpi}))\prod_i |\varpi_i|^{-2}, \qquad F := \frac{\mathbf{1}_{\mathcal{R}'}}{1-\sum_i y_i}.$$

Let $\mathfrak{m}_X$ be the measure on $[c_1,1]$ that gives mass $|\varpi|^{-2}$ to the point $\log|\varpi|^2/\log X$ for each prime ϖ with $|\varpi|^2 \geqslant X^{c_1}$, so that $\sum_{\varpi_1,\dots,\varpi_{K'}} F(\boldsymbol{y}(\boldsymbol{\varpi}))\prod_i |\varpi_i|^{-2} = \int F\, d\mathfrak{m}_X^{\otimes K'}$. By Lemma 17(c) with $a_0 = c_1$, the $\mathfrak{m}_X^{\otimes K'}$-measure of every product of intervals $\prod_i [a_i,b_i)$ tends to $\prod_i \log(b_i/a_i)$, its mass under $\prod_i dy_i/y_i$. The function F is bounded and continuous outside the null set $\partial\mathcal{R}'$, hence Riemann integrable: there are step functions $F_- \leqslant F \leqslant F_+$ on finite grids of such products with $\int (F_+ - F_-)\prod_i dy_i/y_i$ arbitrarily small. This gives (64).

Now consider the left side of (47). For each tuple $\boldsymbol{\varpi}$ in its range, $S(\mathcal{B}_{\varpi_1\cdots\varpi_K},\varpi_K)$ counts the cofactors $t = \varpi_1'\cdots\varpi_h'$ with $\varpi_K \preccurlyeq \varpi_1' \preccurlyeq \cdots \preccurlyeq \varpi_h'$ and $\varpi_1\cdots\varpi_K t \in \mathcal{B}$. Here $t \neq 1$ because $\sum_i y_i \leqslant 1-c_1$, and $h \leqslant 1/c_1$. The tuples with a repeated prime, or with $\varpi_1' = \varpi_K$, number $\ll \sum_{|\varpi'|^2 \geqslant X^{c_1}} X|\varpi'|^{-4} \ll X^{1-c_1}$. The tuples in which two distinct primes have equal norm also number $\ll X^{1-c_1}$. The remaining tuples have distinct norms, and their number is $1/h!$ times the number of such tuples without the order condition on $\varpi_1',\dots,\varpi_h'$. Apply (64) with $K' = K+h-1$ and the free prime $\varpi_0 = \varpi_h'$ to the region

$$\mathcal{R}_h' := \Big\{(\boldsymbol{y},x_1,\dots,x_{h-1}) : \boldsymbol{y}\in\mathcal{R},\ x_i \geqslant y_K,\ 1-\sum_i y_i - \sum_i x_i \geqslant y_K\Big\},$$

where $x_i := \log|\varpi_i'|^2/\log X$ is the exponent of ϖ_i'. Since $|\varpi_1\cdots\varpi_K t|^2 \in [X/2,X)$, the condition $1-\sum_i y_i - \sum_i x_i \geqslant y_K$ of $\mathcal{R}_h'$ is the condition $\varpi_h' \succcurlyeq \varpi_K$ in exponents, up to $\mathcal{O}(1/\log X)$. So the count lies between the values of (64) for the regions obtained from $\mathcal{R}_h'$ by moving this boundary by a fixed small amount in either direction. The integrals over these regions tend to the integral over $\mathcal{R}_h'$ as the amount tends to 0. The integrand $\prod_{i\leqslant h} 1/x_i$, with $x_h = w - \sum_{i<h} x_i$ and $w = 1-\sum_i y_i$, is symmetric in the h coordinates, so $1/h!$ of the integral over $\mathcal{R}_h'$ is the integral over the subset with $x_1 \leqslant \ldots \leqslant x_h$. Hence the left side of (47) is

$$\frac{X}{2\log X}\Big(\int_{\mathcal{R}} g\Big(1-\sum_i y_i;\, y_K\Big)\frac{d\boldsymbol{y}}{y_1\cdots y_K} + o(1)\Big),$$

where

$$g(w;b) := \sum_{h\geqslant 1}\int_{b\leqslant x_1\leqslant\ldots\leqslant x_{h-1}\leqslant w-\sum_{i<h}x_i}\prod_{i<h}\frac{dx_i}{x_i}\cdot\Big(w-\sum_{i<h}x_i\Big)^{-1}.$$

It remains to show that $g(w;b) = \omega(w/b)/b$, since then the integrand $g(1-\sum_i y_i; y_K)/(y_1\cdots y_K)$ is that of (47). The term $h=1$ of $g(w;b)$ is $1/w$ for $w \geqslant b$. For $h \geqslant 2$, the smallest coordinate $x = x_1$ lies in $[b, w/2]$, and the other coordinates $x_2 \leqslant \ldots \leqslant x_h$ are at least x and sum to $w-x$, so they give the term $h-1$ of $g(w-x;x)$. Hence

$$g(w;b) = \frac{1}{w} + \int_b^{w/2} g(w-x;x)\,\frac{dx}{x} \quad (w\geqslant b), \qquad g(w;b) = 0 \quad (w<b). \tag{65}$$

For $w \geqslant b$, put $\omega(w/b)/b$ in place of g on the right side of (65). The substitution $x = w/(t+1)$ and then (19) at $u = w/b$ give

$$\frac{1}{w} + \int_b^{w/2} \omega\Big(\frac{w-x}{x}\Big)\frac{dx}{x^2} = \frac{1}{w}\Big(1+\int_1^{w/b-1}\omega(t)\,dt\Big) = \frac{1}{b}\,\omega\Big(\frac{w}{b}\Big),$$

so $\omega(w/b)/b$ satisfies (65). Both functions vanish for $w<b$, and (65) determines them by induction on $\lceil w/b\rceil$, since $(w-x)/x \leqslant w/b - 1$. So $g(w;b) = \omega(w/b)/b$, and (47) follows from (46). □

D Inequalities used in the decomposition

Table 4 lists the inequalities between the parameters (4) that the decomposition uses, directly or through the lemmas of Sections 4–6. The exponent $\chi = (\nu+2\theta)/2$ in the table is that of Section 7. Each inequality of a row is equivalent to the positivity of the quantity of that row. The quantities are linear in θ and non-negative at $\theta = 1/9$ and $\theta = 1/8$, hence positive for $1/9 < \theta < 1/8$.

Quantity	Inequality and use	Value at $\theta = 0.1129$
$\theta - 1/10$	$3\sigma/2 < \tau$, $\sigma - 3\theta < \theta$; Lemmas 12 and 13	0.0129000
$1/8 - \theta$	$\tau/5 < \sigma - 2\theta$; Lemma 13	0.0121000
$(1 - 8\theta)/3$	$\nu + \rho < 1$; Lemma 9	0.0322667
$1/6 - \theta$	$2\theta < \sigma$; Lemmas 2 and 9	0.0537667
$2\theta - 1/6$	$\sigma - \theta < 1/3$; $j = 0$ in Range 2	0.0591333
$3\theta/2 - 1/12$	$\sigma < (1/3 + \tau)/2$; $PQ \geqslant W$ in Range 2 and on R	0.0860167
$(3 - 22\theta)/12$	$2\zeta - 1/3 < \nu$; $PQ^2 \leqslant X^\nu$ in Range 2	0.0430167
$(1 + 6\theta)/12$	$\zeta - 1/3 < \rho$; $Q \leqslant X^\rho$ in Range 2	0.1397833
$(1 - 2\theta)/6$	$\zeta < \tau$; $PQ \leqslant X^{\tau-\varepsilon}$ where $\alpha + \beta \leqslant \zeta$	0.1290333
$(14\theta - 1)/6$	$\sigma < \zeta$, $\zeta < 2\rho$; $P \leqslant X^\zeta$ on R, $Q \leqslant X^\rho$ in Range 3	0.0967667
$(5 - 34\theta)/24$	$\zeta - \sigma < \rho/2$, $\rho < \sigma$; $\mathsf{R} \neq \emptyset$, Lemma 9	0.0483917
$(26\theta - 1)/24$	$\sigma + \rho/2 < \tau$; $PQ \leqslant X^{\tau-\varepsilon}$ on R	0.0806417
$(1 - 7\theta)/3$	$\tau/3 < \sigma - \theta$; Lemma 13	0.0699000
$(46\theta - 5)/12$	$\sigma - \theta < \rho$; $j = 0$ in Range 3, $Q \leqslant X^\rho$ in Range 5	0.0161167
$(58\theta - 5)/24$	$\sigma < (\rho + \tau)/2$; $PQ \geqslant W$ in Range 3	0.0645083
$(1 - 2\theta)/24$	$\chi < \zeta$; Lemma 9 in Range 1	0.0322583
$(11 - 82\theta)/36$	$\nu/3 < \sigma - \theta$; $\mathsf{D}_8 \neq \emptyset$	0.0483944
$(46\theta - 5)/24$	$(\nu + \sigma - \theta)/2 < \zeta$; $PQ \leqslant X^\zeta$ in Range 5	0.0080583
$5\theta/3 - 1/6$	$\sigma < 2\tau/3$; $PQ \geqslant W$ on E_9	0.0215000
$5\theta/2 - 1/4$	$\sigma - \theta < \tau/2$; E_{10} lies in Range 5	0.0322500
$17\theta/5 - 3/10$	$\sigma - 2\theta < 2\tau/5$; $\lvert pq\rvert^2 \geqslant WX^{-2\theta}$ in Range 6	0.0838600
$21\theta/5 - 2/5$	$\sigma - 3\theta < \tau/5$; $j(X^\alpha) \leqslant 2$ in Range 6	0.0741800

Table 4: Inequalities between the parameters (4), written as quantities that must be positive, with the place where each is used and its value at $\theta = 0.1129$.

E Comparison with the argument of Harman and Lewis

Table 5 lists the points where this paper adds to, or differs from, the argument of Harman and Lewis [10]. The paragraphs and remarks after the table give the details, in the order of the table.

Remark 24 *Harman and Lewis print, in their (27) and (28), the conditions $ut \in \mathcal{A}$ and $uvt \in \mathcal{A}$, here written in the notation of Lemma 23. The proof of their Lemma 6 selects the norm range by the factor $(X/|u|^2)^s(1-2^{-s})$ in a Perron integral of the Hecke L-function $L(s, \lambda^m)$ of Harman and Lewis [10, (19)], with no condition on the argument, and the proof of their Lemma 7 proceeds in the same way. So the conditions are $ut \in \mathcal{B}$ and $uvt \in \mathcal{B}$, which is also the form needed in Lemma 20. Their stated bound is $X\eta$, for their small parameter η. This bound holds for every $\eta = (\log X)^{-A}$, since the proofs save $\exp(-c(\log X)^{1/10})$ or a power of μ. The hypothesis $\theta > 1/14$ of their Lemma 7 holds throughout this paper.*

Remark 25 *Harman and Lewis [10, Lemma 5 and (30)] state the bilinear estimate for $U, V \geqslant W$ and $UV \leqslant X\mu^{-1}$, with the prime ϖ running over all norms up to $4X/(UV)$. Their proof applies their Lemma 4, which is Lemma 22, with $|t| \leqslant X$ and $S = 4X/(UV)$. As stated, Lemma 22 then requires*

$$S \geqslant \exp\bigl(((1+\sigma)\log X + A\log\log X)^{4/5}\bigr),$$

and this lower bound is about $\mu^{1.3}$. This requirement is not met when UV is close to $X\mu^{-1}$. Lemma 2 stays within the range of Lemma 22: its prime has norm at least $\mu^4/2$. Every use of the bilinear estimate in this paper goes through Lemma 2, and primes of norm below μ^4 are handled by the fundamental lemma, Lemma 3.

Harman and Lewis invoke a fundamental lemma of sieve theory in the proofs of their Lemmas 8, 9, 11 and 12. Lemma 3 states the form used in this paper, at levels up to μ^4, and Appendix B proves it by Brun's pure sieve.

Item of Harman and Lewis [10]	Difference	Treated in
(27), (28)	conditions $ut \in \mathcal{A}$, $uvt \in \mathcal{A}$ read as $\mathcal{B}$	Remark 24
Lemma 5 and (30)	prime restricted to norms at least $\mu^4/2$, within the range of their Lemma 4 (Lemma 22)	Remark 25
fundamental lemma in Lemmas 8, 9, 11, 12	stated and proved here	Lemma 3
Lemmas 8–10	proofs given in full	Lemmas 4, 5, Corollary 6
Lemmas 11, 12, (30)	conditions coupling variables removed by short intervals	Lemma 7
Lemma 11	lower bound X^τ/P for Q not needed	Lemma 9
Lemma 12	hypothesis $R \geqslant X^\theta$ not needed	Lemma 10
Lemma 13	a level not of the form $R\|n\|^k$; case (b) added	Lemma 11
part 6 of §5	step stated and proved here	Lemma 12
(37), sum of $S(\mathcal{E}_p, z(\|p\|^2))$	level above the prime for small $\|p\|^2$	Lemma 13, Remark 26
§5	decomposition extended to $\theta < 3/26$	Section 7
regions of φ_2 and φ_{10}, value of χ	printed limits and value made precise	Remark 27
cuts in φ_3, φ_7, φ_9	the cut in φ_7 lies in G; that in φ_3 and φ_9 follows from the limits of δ	Section 8; (42) and (18)
bounds for φ_2, φ_5, φ_9 at $\theta = 0.119$	below the losses	Table 6
(17)	factor $1+\mathcal{O}(\eta)$ replaced by the term $o(1)$	Lemma 15

Table 5: Points where this paper adds to, or differs from, the argument of Harman and Lewis [10], and the place where each is treated. The column "Item of Harman and Lewis [10]" uses the numbering of lemmas and displays of Harman and Lewis. The symbols η and φ_k are those of Harman and Lewis [10], and χ is the exponent of Section 7. The cuts are the inequalities by which Harman and Lewis remove subregions of the regions of φ_3, φ_7 and φ_9.

Lemmas 8, 9 and 10 of Harman and Lewis [10] give asymptotic formulas for $\sum_n a_n S(\mathcal{E}_n, l(n))$, and Lemmas 4 and 5 and Corollary 6 prove them. The proofs of Harman and Lewis apply the bilinear estimate with a prime of norm down to μ, while Lemma 2 keeps the prime at norm at least $\mu^4/2$. For the prime q of the first Buchstab step, they remove the condition $|q|^2 < l(n)$ by Perron's formula [10, p. 128], and the proof of Lemma 4 removes it by short intervals. Their Lemma 9 iterates Buchstab's identity (6) up to $(\log X)^{1/5}$ times. The proof of Lemma 5 organizes this iteration into stages, with coefficients bounded uniformly in the stage.

Lemmas 8 and 9 of Harman and Lewis [10] concern S and coefficients a_n bounded by a divisor function, while Lemmas 4 and 5 are stated for S' and for $a_n \geqslant 0$. The forms with S and with complex a_n follow. For a divisor-bounded $a_n \geqslant 0$ and a function l,

$$\sum_n a_n S(\mathcal{E}_n, l(n)) = \sum_n a'_n S'(\mathcal{E}_n, l(n)),$$

where $a'_n := a_n$ if every prime factor of n has norm at least $l(n)$, and $a'_n := 0$ otherwise. A divisor-bounded complex sequence is a combination of 4 non-negative divisor-bounded sequences.

Lemma 8 of Harman and Lewis [10] is the case of Lemma 4 with $n \sim Y$ and $l(n) = R|n|^k$, where $-2 \leqslant k \leqslant 2$ and $\mu \leqslant l(n)$.

Lemma 9 of Harman and Lewis [10], for $n \sim Y$ with $X^\theta \ll Y \ll W$ and $l(n) = R|n|^k \ll z(Y)$, follows from Lemmas 5 and 4. For the n with $|n|^2 < W$, Lemma 5 applies, since then $z(|n|^2) \geqslant z(Y)/2$. For the n with $|n|^2 \geqslant W$, Lemma 4 applies, since $z(Y) \leqslant X^\tau/Y \leqslant 2X^\tau/|n|^2$. Lemma 10 of Harman and Lewis [10] is Corollary 6.

In the proofs of their Lemmas 11 and 12, Harman and Lewis remove the conditions that couple the variables before applying their bilinear estimate (30), by "applying Perron's formula to disentangle the

relationship between r and s" and by "removing the interdependence between s and t" [10, p. 130]. Lemma 7 removes them by short intervals.

In their Lemma 11, Harman and Lewis assume

$$W \leqslant P \ll X^{\chi}, \qquad \max(X^{\tau}/P, PX^{-2\theta}) \ll Q \ll X^{\nu}/(P\mu), \qquad \chi = (34\theta+7)/24 = (\nu+2\theta)/2.$$

Since $\chi < \zeta$, their hypotheses imply those of Lemma 9 on boxes at distance at least ε from the boundary of the region that these hypotheses define. Their lower bound X^{τ}/P for Q is not used.

Lemma 12 of Harman and Lewis [10] is case (a) of Lemma 10 with $\mathcal{Q} = [Q, 2Q)$ and $\mathcal{I} = [R, 2R)$ for $R \geqslant X^{\tau}/(PQ)$, without the constraints $q \prec p$ and $r \prec q$. The proof of Lemma 10 applies without the constraint $r \prec q$, and, when $P \geqslant \inf \mathcal{Q}$, without the constraint $q \prec p$, since dropping a constraint changes none of the checks in that proof. Their hypothesis $R \geqslant X^{\theta}$ is not used in the proof of Lemma 10. In their Lemma 13, where their Lemma 12 is applied, $R \geqslant X^{\tau}/(PQ) \geqslant X^{\tau-\zeta} > X^{\theta}$ holds automatically. In the decomposition of Section 7, the inequality $R \geqslant X^{\theta}$ does not always hold.

Lemma 13 of Harman and Lewis [10] is case (a) of Lemma 11. Their condition "$Q^2P < X^{\tau}$" is the case $S_{\geqslant} = 0$. Case (a) of Lemma 11 also needs $Q \leqslant X^{\rho-\varepsilon}$, which is not among their hypotheses. Their hypotheses imply it for small ε, because $Q^2 \leqslant 2PQ \leqslant 2X^{\zeta}$ and $\rho - \zeta/2 = (14\theta-1)/12 > 0$. Their proof cites their Lemmas 8 and 12. The sum over the pairs with $|q|^2 < X^{\tau}/(PQ)$ has the level $|q|^2$, which is not of the form $R|n|^k$ of their Lemma 8 for $n = pq$. Lemmas 4 and 7 of this paper allow this level. Case (b) of Lemma 11 has no counterpart in their Lemma 13. Section 3 describes its use on R.

Part 6 of Harman and Lewis [10, §5] uses the case $j = 1$ of Lemma 12, which is stated and proved here. For $j = 0$, Lemma 12 is Lemma 14 and (36) of Harman and Lewis [10], and their condition $PQ^3 < X^{\tau}$ is the case $S_{\geqslant} = 0$.

Remark 26 *Harman and Lewis [10, (37)] sum $S(\mathcal{E}_p, z(|p|^2))$ over all p with $X^{\theta} \leqslant |p|^2 < X^{1/2}$. For $p \in \Pi_2$ the function $S(\mathcal{E}_p, z(|p|^2))$ sifts p itself and vanishes, while the term $S(\mathcal{E}_p, p)$ of the first Buchstab step remains. Identity (12) keeps the sum of $S(\mathcal{E}_p, p)$ over $p \in \Pi_2$, and Lemma 13 gives it an asymptotic formula.*

The primes with $\tau/4 \leqslant \alpha < \sigma - 2\theta$ exist only for $\theta < 3/26$, so they do not occur in the decomposition of Harman and Lewis [10, §5] at $\theta = 0.119$. Section 7 treats them in the paragraph after (42).

Remark 27 *The regions of Section 7 are derived from the lemmas. Two of them differ from the regions that Harman and Lewis print for φ_2 and φ_{10}. Harman and Lewis [10, p. 132] print the lower limit $\zeta - \alpha$ for β in φ_2. For $\alpha > (1+10\theta)/6$, this limit lies below $\beta_0(\alpha) = (\tau - \alpha)/2$, and the strip between these limits contains no pair of $\sum^*$. Harman and Lewis print the upper limit $1/2 - 2\theta - \alpha$ for β in φ_{10}, which makes the region empty for every $\alpha \geqslant \tau/4$. The condition $|pq|^2 < W$ of Lemma 12 with $j = 0$ gives the upper limit $\sigma - \alpha$. They also print $\chi = 0.46205$ at $\theta = 0.119$, while $(34\theta+7)/24 = 0.46025$.*

Harman and Lewis discard the third sums of (29) and (31), except on a few subregions of the regions of φ_3, φ_7 and φ_9 that they remove by further inequalities, the *cuts*. Harman and Lewis [10, p. 133] use a grouping of the kind that defines G and L only in their φ_7. There, the condition $\delta > 1 - \alpha - \gamma - \sigma$ when $\alpha + \gamma \geqslant \sigma$ excludes from the region of φ_7 the subset of G given by $\mathcal{S} = \{p, r\}$. This exclusion changes no loss once G is added, and the evaluation of the decomposition of Harman and Lewis in Section 10 keeps it. Their cut $\gamma \leqslant \alpha + \beta - 2\theta$ in φ_3 and φ_9 is implied by the limits of δ. On E_3 and E_9, $\delta \geqslant \tau - \alpha - \beta$ by (42), and the integrand of $\Phi(\mathsf{F}')$ in (18) vanishes unless $\delta \leqslant (1 - \alpha - \beta - \gamma)/2$. These bounds are compatible only if $\gamma \leqslant 1 - 2\tau + \alpha + \beta = \alpha + \beta - 2\theta$.

At $\theta = 0.119$, the bounds that Harman and Lewis print for φ_2, φ_5 and φ_9 lie below the losses of Table 6.

Harman and Lewis [10, (17)] state the case $K = 2$ of Lemma 15 with a factor $1 + \mathcal{O}(\eta)$ in place of the term $o(1)$ of (47), where $\eta = (\log X)^{-A}$ as in Remark 24. For $K = 4$, Lemma 15 is the form they use for φ_3, φ_5, φ_7, φ_9, φ_{10} and φ_{11}. The passage from the prime number theorem to the integral already loses $\mathcal{O}(1/\log X)$ through Lemma 17(c), so the factor $1 + \mathcal{O}(\eta)$ is not available in general.

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Loss	Region	First method	Second method	Bound of Harman and Lewis
φ_1	D_1	0.2747838	0.2747838	0.275
φ_2	$\mathsf{D}_2 \cup \mathsf{R}$	0.3456458	0.3456458	0.341
φ_3	E_3	0.0000918	0.0000918	0.001
φ_4	D_4	0.1554561	0.1554561	0.156
φ_5	E_5	0.0011859	0.0011859	0.001
φ_6	D_6	0.1116255	0.1116255	0.112
φ_7	E_7	0.0008847	0.0008847	0.001
φ_8	D_8	0.0356394	0.0356394	0.036
φ_9	E_9	0.0010752	0.0010752	0.001
φ_{10}	E_{10}	0.0161420	0.0161419	0.017
φ_{11}	E_{11}	0.0000285	0.0000285	0.001
total		0.9425585	0.9425585	0.943

Table 6: The losses (18) of the decomposition of Harman and Lewis at $\theta = 0.119$, computed by the methods of Section 10 with the regions derived from the lemmas (Remark 27) and with the cut of Harman and Lewis in φ_7, rounded to 7 decimal places. The column "Bound of Harman and Lewis" gives the bounds of Harman and Lewis [10, pp. 132–134].

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